

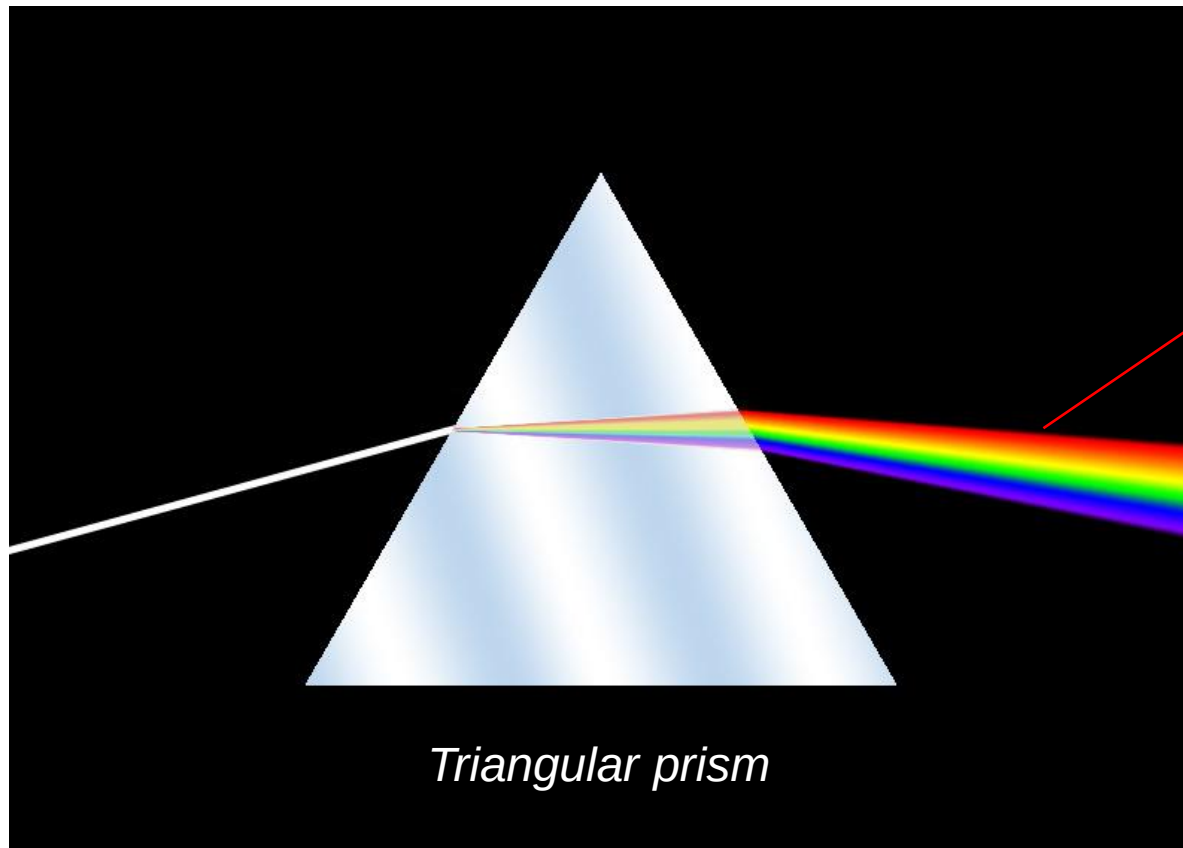


Digital Image Processing

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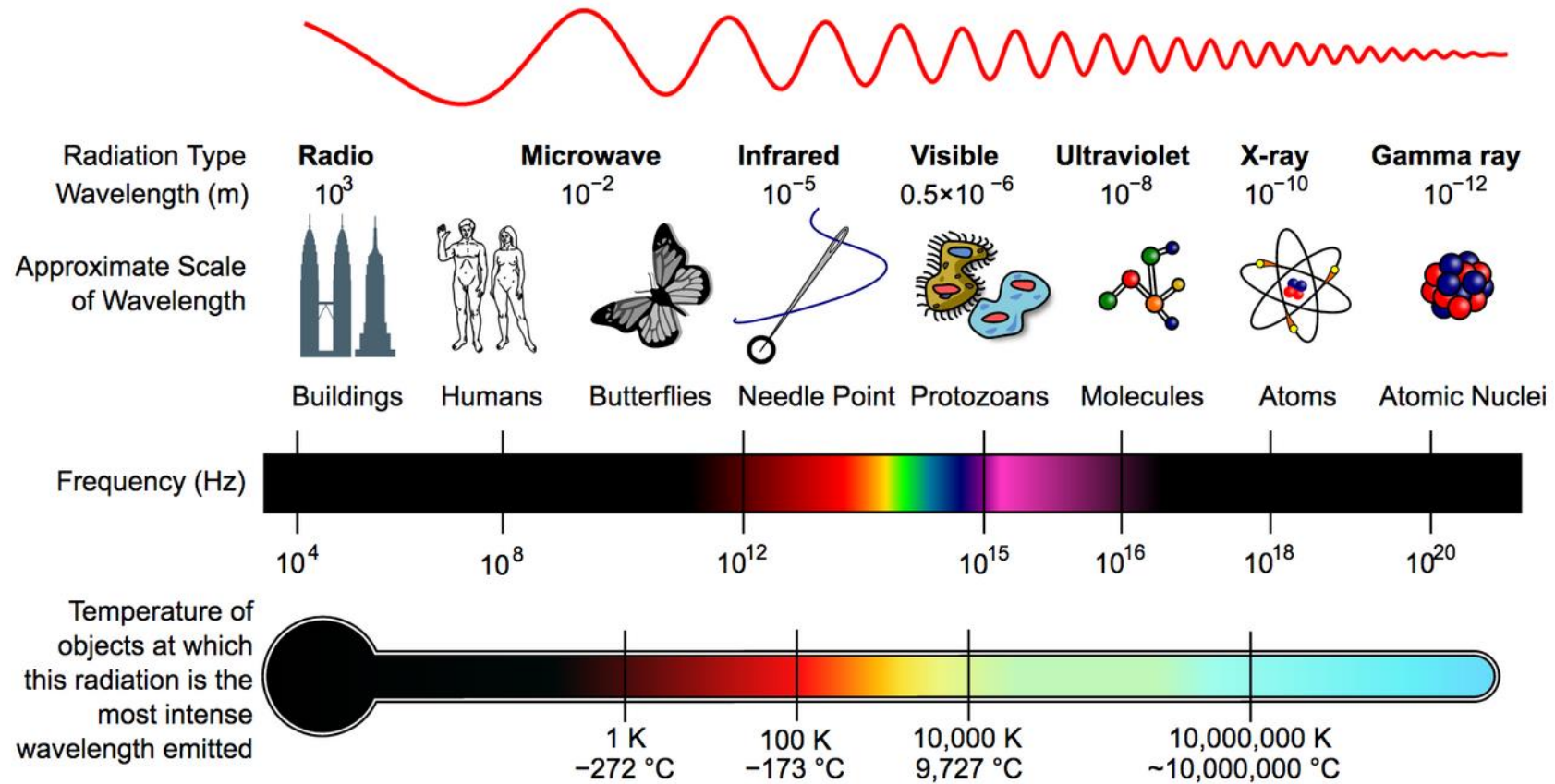
Color



Spectral components of light

The prism deflects the light with different wavelengths at different angles

Electromagnetic Spectrum



Color Recreation

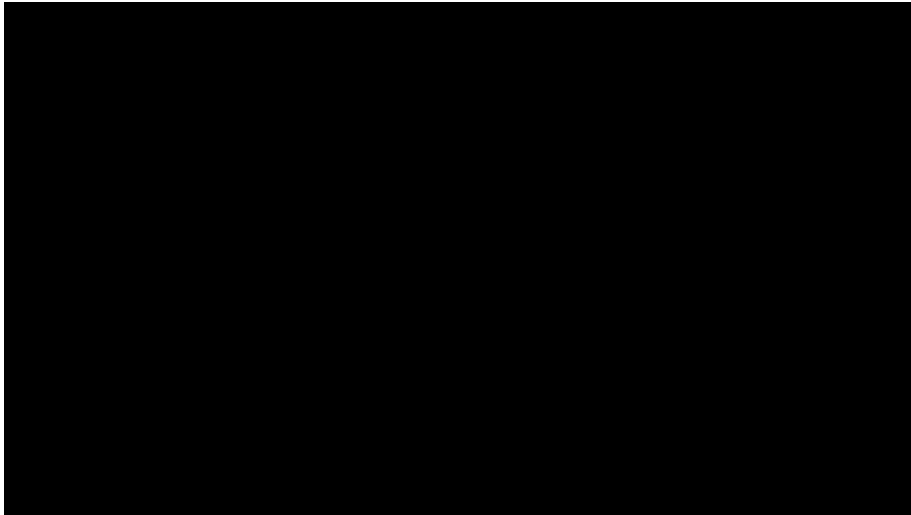
- ☐ Why can a display or projector show various colors?
 - ☐ Additive color mixing - combining different intensities of primary colors
 - ☐ Each pixel or light source in a display or projector adjusts the brightness of the three colors independently.
 - ☐ Projectors use separate light sources for each color or filter white light to achieve this.
- ☐ Does a projector need to emit light with distinct wavelengths for all the colors it shows?
 - ☐ No
 - ☐ It typically uses light sources for the three primary colors (red, green, and blue) with distinct wavelengths

色彩顯示的核心不是精確模仿自然光譜，而是利用人眼的視覺特性（對 RGB 敏感）來模擬出所有色彩。

Additivity and Subtractivity of Color Mixing

LIGHT MIXING (ADDITIVE)

加法混色是通過將不同顏色的光疊加在一起來產生新顏色的過程。當所有顏色光完全疊加時，結果趨向於白色光。



Projector

PLAY-DOH (SUBTRACTIVE)

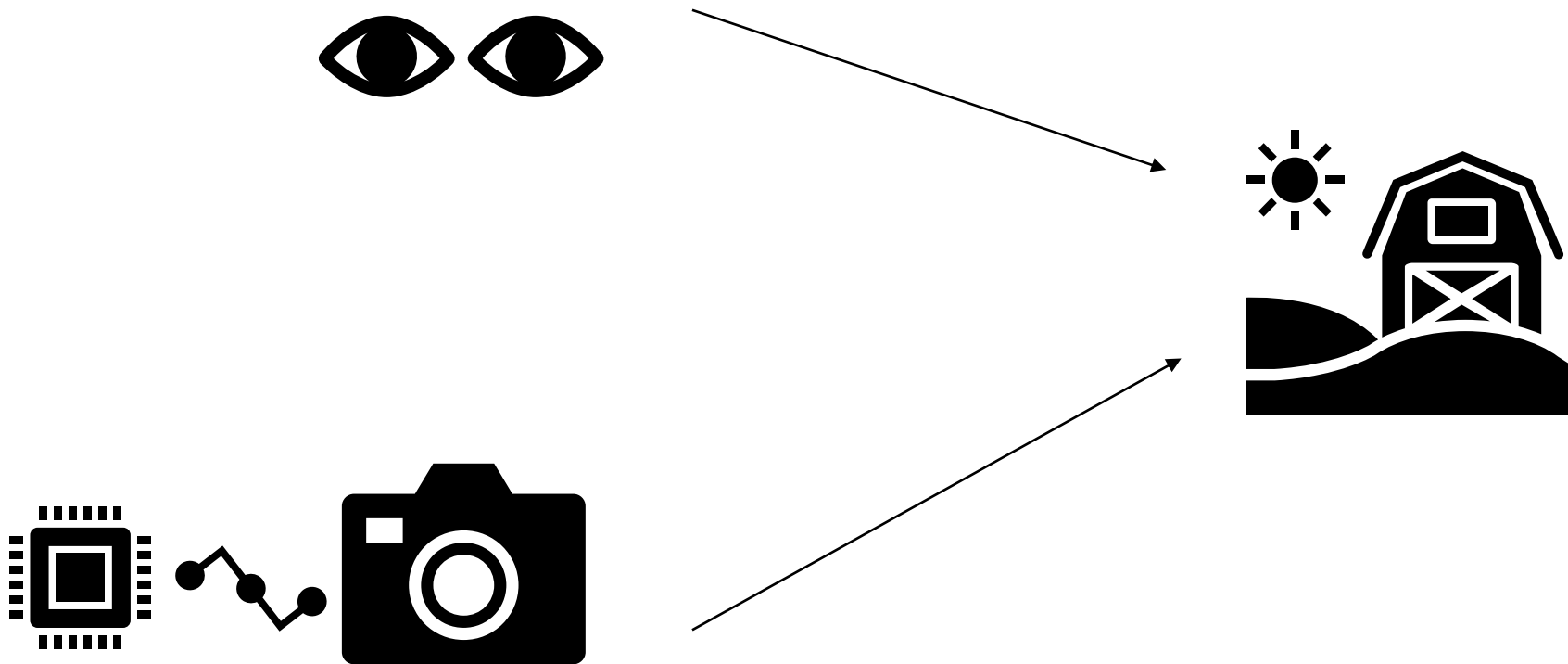
減法混色是通過將不同顏色的顏料或染料混合，從白光中吸收（減去）特定波長的光來產生新顏色的過程。當所有顏色混合到極致時，結果趨向於黑色。



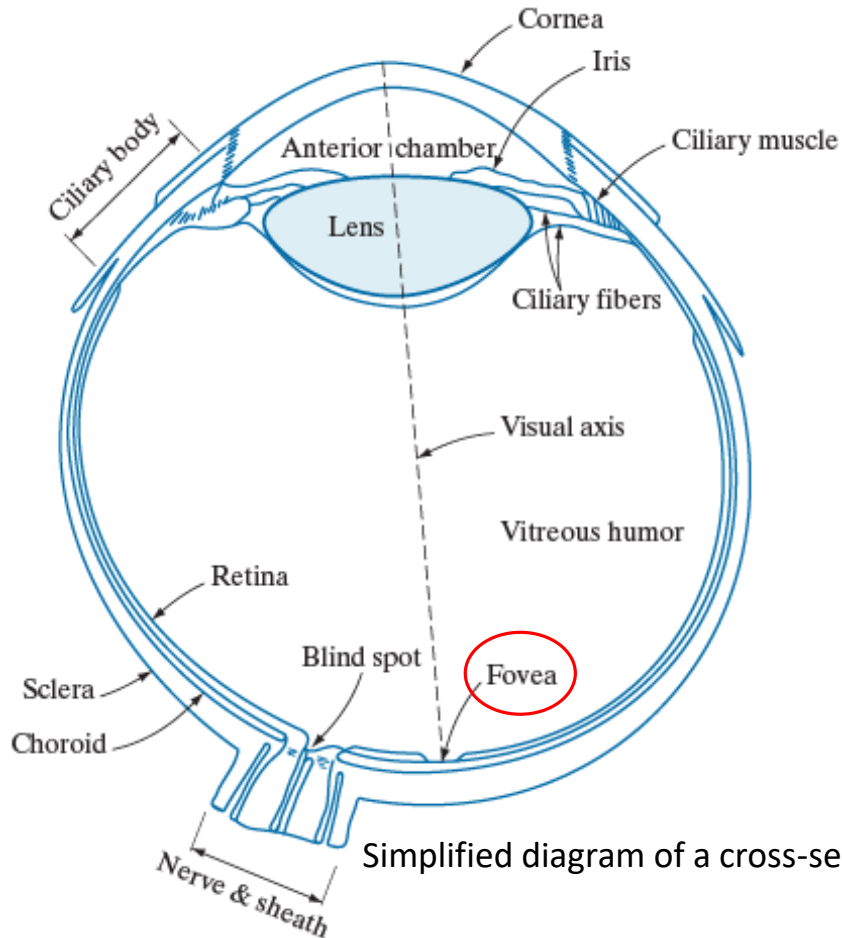
Image Processing: Extracting Useful Information

- ❑ **Understanding the Human Visual System** is key to developing techniques that help humans and machines gain meaningful insights from images.
- ❑ **Human Eye vs. Camera:**
 - ❑ Differences in image capture, light sensitivity, color perception, and dynamic range.
- ❑ **Human Visual System:**
 - ❑ How the brain processes visual data — depth, color, focus — to interpret images.
- ❑ *This foundation guides the creation of image processing methods tailored for human perception and machine analysis.*

Human Vision vs. Machine Vision



Human Eye vs. Camera



Retina are full of sensors:

1. Cone cells
2. Rod cells

Term	Chinese	Corresp. To Camera
Cornea	角膜	鏡頭
Iris/pupil	虹膜/瞳孔	光圈
Retina	視網膜	底片
Lens	水晶體	對焦裝置
Eyelid	眼瞼	鏡頭蓋

Simplified diagram of a cross-section of the human eye

Retina: The Light-Sensitive Layer Full of Sensors

- ❑ The **retina** contains specialized photoreceptor cells that convert light into neural signals, enabling vision.

- ❑ **Cone Cells**

- ❑ **Function:** Responsible for **color vision** and **sharp detail**

- ❑ Works Best: In **bright light** (photopic vision). **明視 (photopic vision)**

- ❑ **Location:** Concentrated in the **fovea** (central retina).

- ❑ **Types:**

- ❑ **S-Cones** (Short) – Blue light
 - ❑ **M-Cones** (Medium) – Green light
 - ❑ **L-Cones** (Long) – Red light

中央凹 (fovea)

這解釋了：

為什麼在暗處我們只看到黑白、不見顏色
為什麼我們的中央視野比周邊清楚

- ❑ **Rod Cells**

- ❑ **Function:** Enable vision in low light and peripheral vision.

- ❑ **Location:** Spread throughout the **peripheral retina**, absent in the fovea.

- ❑ **Sensitivity:** Highly sensitive to light, but **do not detect color**.

- ❑ **Works Best:** In **dim light** (scotopic vision).

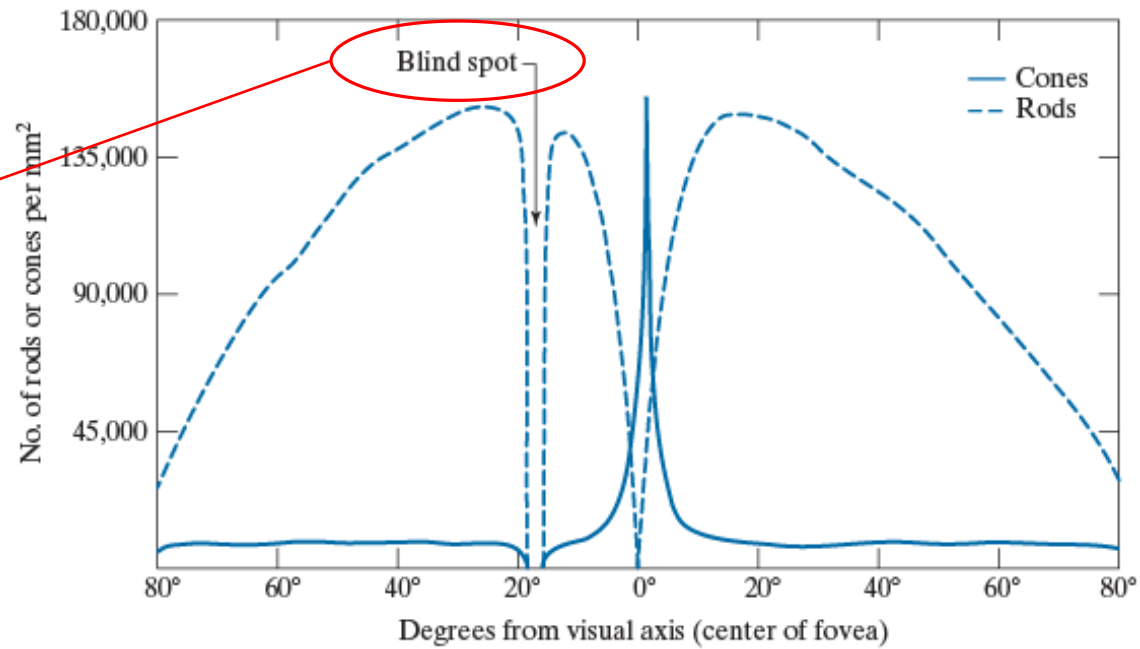
“dim light” (昏暗光線)

暗視 (scotopic vision)

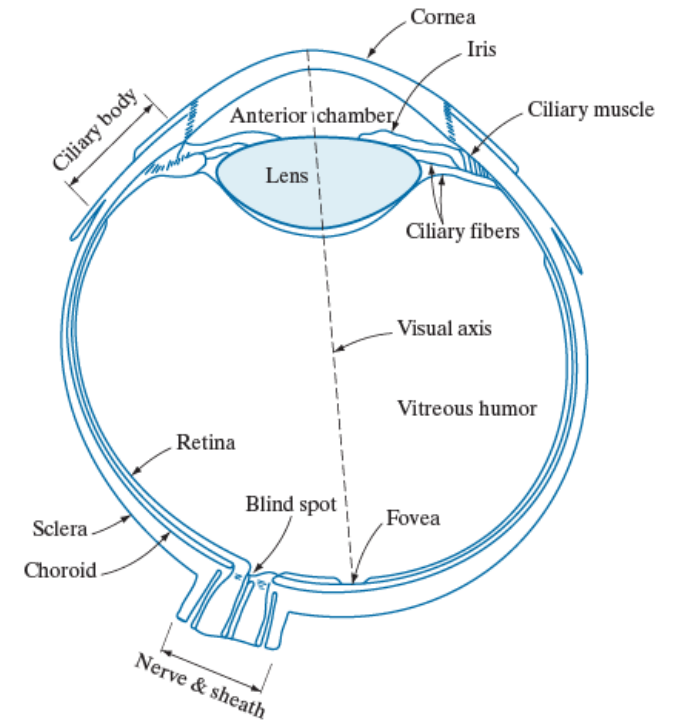
Fovea: The Center of Sharp Vision

- ❑ The **fovea** is a small **pit** in the **macula of the retina**, responsible for **sharp central vision** essential for tasks like reading, driving, and recognizing faces.
- ❑  **High Cone Density:** Packed with cone photoreceptors for **color vision** and **fine detail**, with no rod cells.
- ❑  **Color & Detail Perception:** Enables the **sharpest**, most vibrant vision in the center of our gaze.
- ❑  **Visual Acuity:** **Light focuses on the fovea** when looking directly at an object, providing **maximum clarity**.
- ❑ The fovea is the key to seeing vivid colors and fine details in our direct line of sight.

Can't see anything



Distribution of rods and cones in the retina



Visual Adaptation Curve: Scotopic vs. Photopic Vision

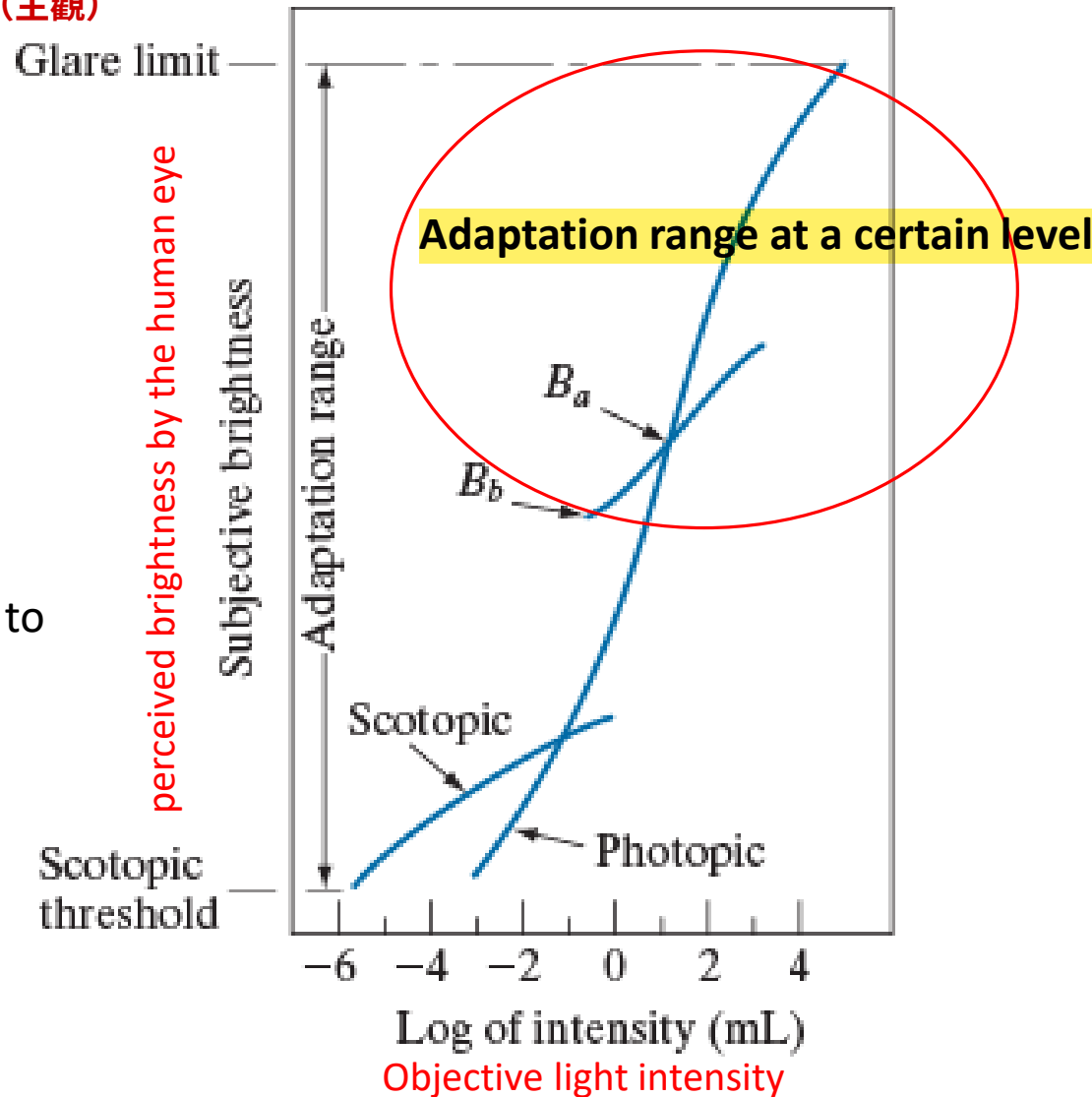
暗視 (Scotopic)、明視 (Photopic) 和中間視 (Mesopic)

橫軸：實際光強度 (對數刻度)
縱軸：人眼感受到的亮度 (主觀)

FIGURE 2.4

Range of subjective brightness sensations showing a particular adaptation level, B_a .

Human eyes adapt to different light intensities



Photopic Vision (Cone-Dominated) ☀️:

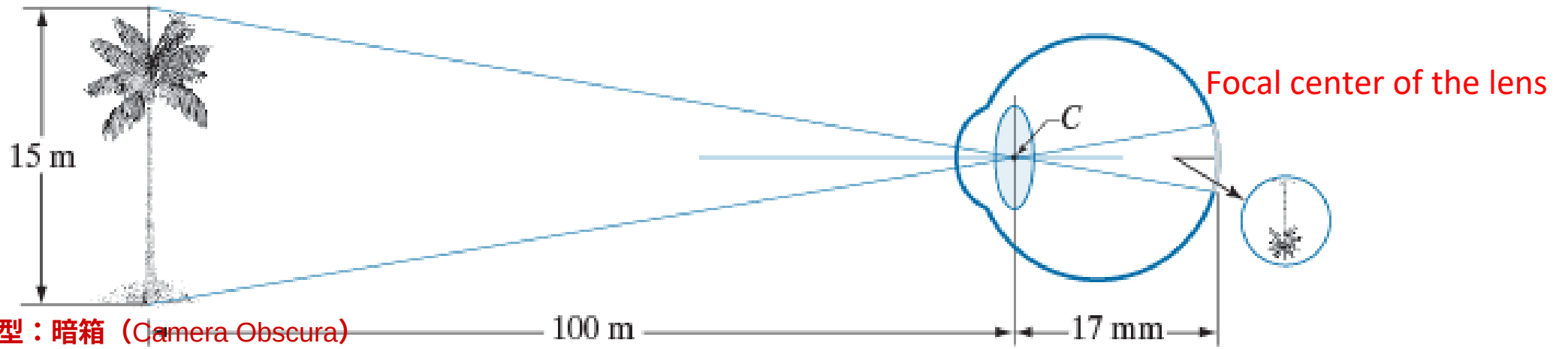
- Active in **bright-light** conditions.
- Responsible for **sharp, color vision**.
- Less sensitive to dim light compared to rods.

Mesopic Region (Transition Zone) 🌅:

- Overlap between scotopic and photopic ranges.
 - Both rods and cones are active.
- Represents dawn, dusk, or dimly lit environments.

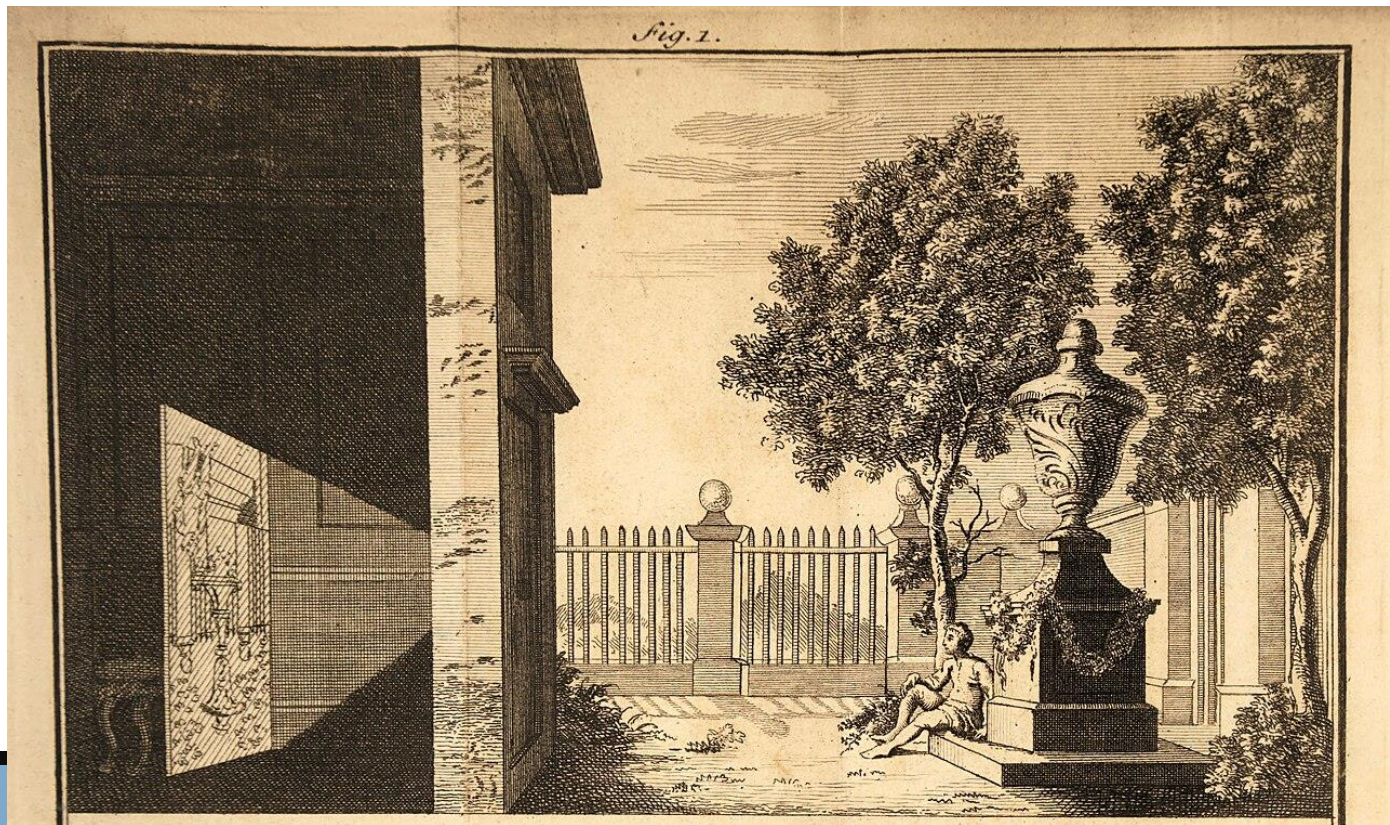
Scotopic Vision (Rod-Dominated) 🌙:

- Active in **low-light** conditions.
- Scotopic threshold**: The minimum light level rods can detect.
- High sensitivity to light but **no color perception**.



早期的相機模型：暗箱 (Camera Obscura)

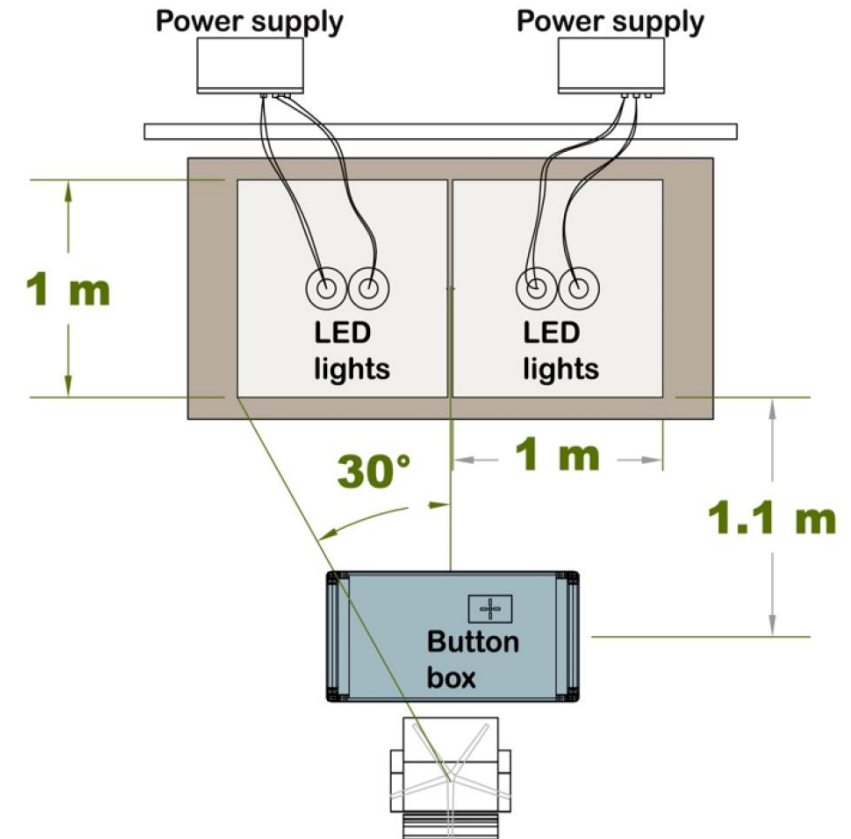
Camera obscura

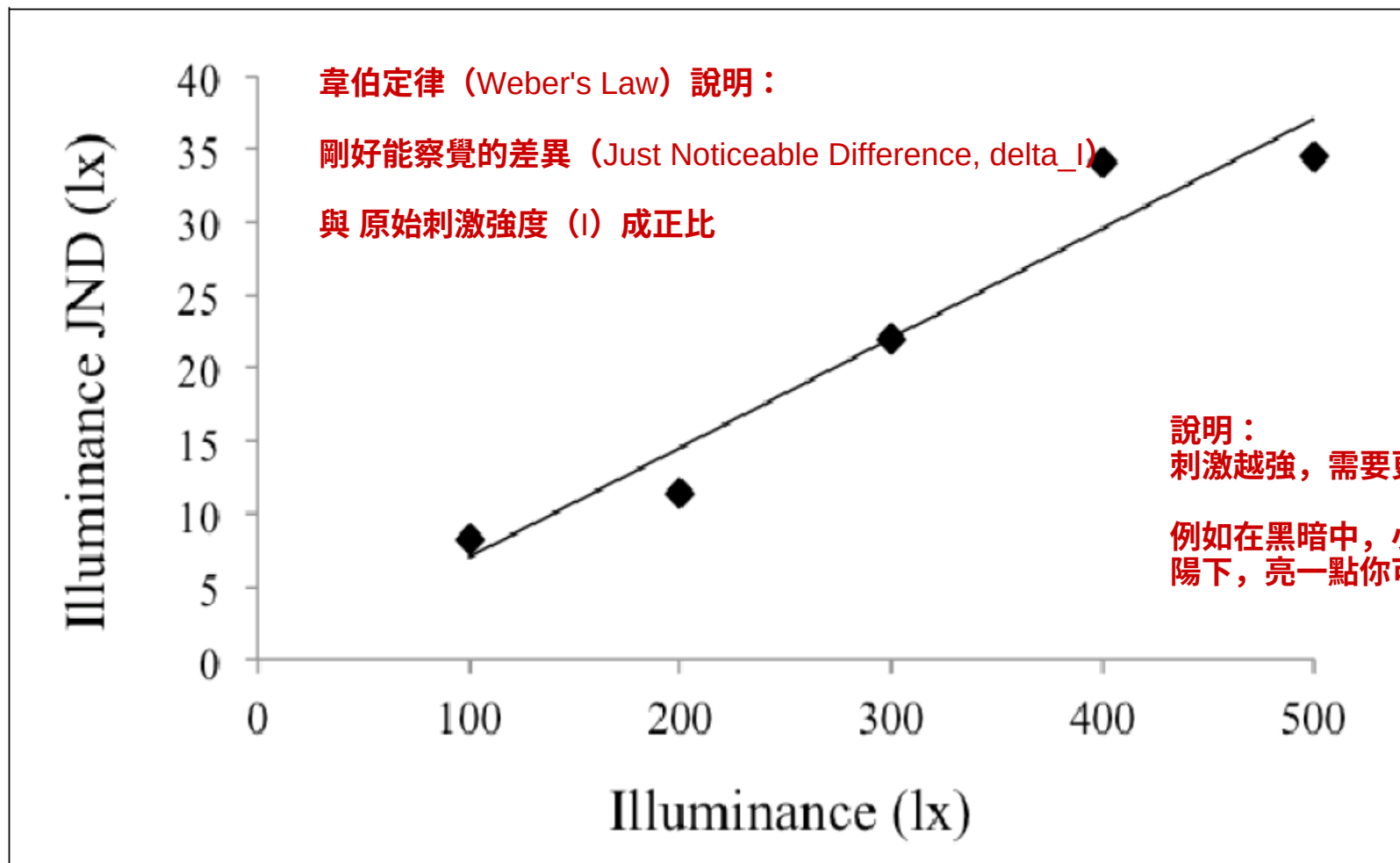


<https://www.cbsnews.com/pictures/abelardo-morells-stunning-camera-obscura-images/6/>

Weber's law & Weber's ratio for human vision

Experiment set to determine the capability of the human visual system for brightness discrimination





有一半的機會判斷正確。受試者在50%的情況下能辨別出的最小光強度增量（即剛好能感知的閾值）。

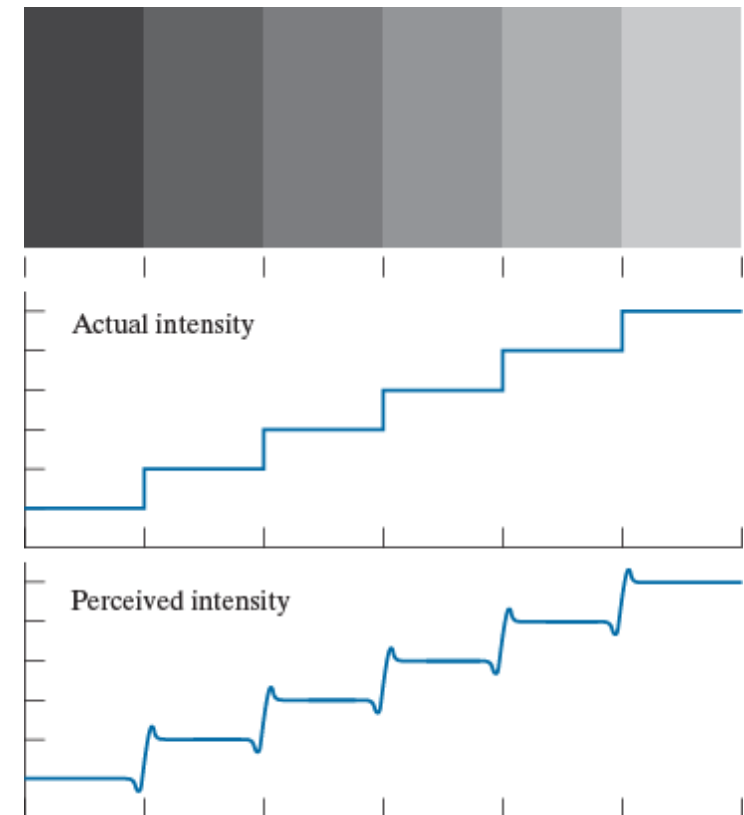
Weber's law states that the **just-noticeable difference (ΔI)** between two stimuli is proportional to the initial stimulus intensity (I), expressed as $\Delta I/I = c$, where c is a constant. As stimulus intensity increases, the required difference to notice a change also increases proportionally.

(Ref: [Britannica - Weber's Law](#))

Human Visual Perception & Surrounding Influence

- ❑ Our **visual system interprets objects relative to their environment**, affecting brightness, contrast, and color perception
- ❑ Weber's law
 - ❑ The **just-noticeable difference (ΔI)** between two stimuli is **proportional** to the initial stimulus intensity (I)
- ❑ Mach band effect **馬赫帶效應**
 - ❑ **Edge enhancement phenomenon** where **gradual changes in brightness appear exaggerated** at boundaries.

在亮暗交界處，邊界的變化會被「誇大」

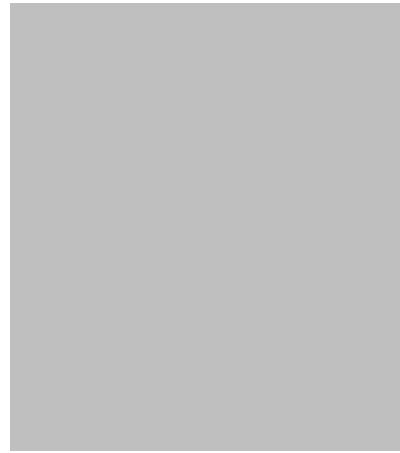


Color Illusions

- The human eyes are tricked by surrounding colors into incorrect interpretation of color.

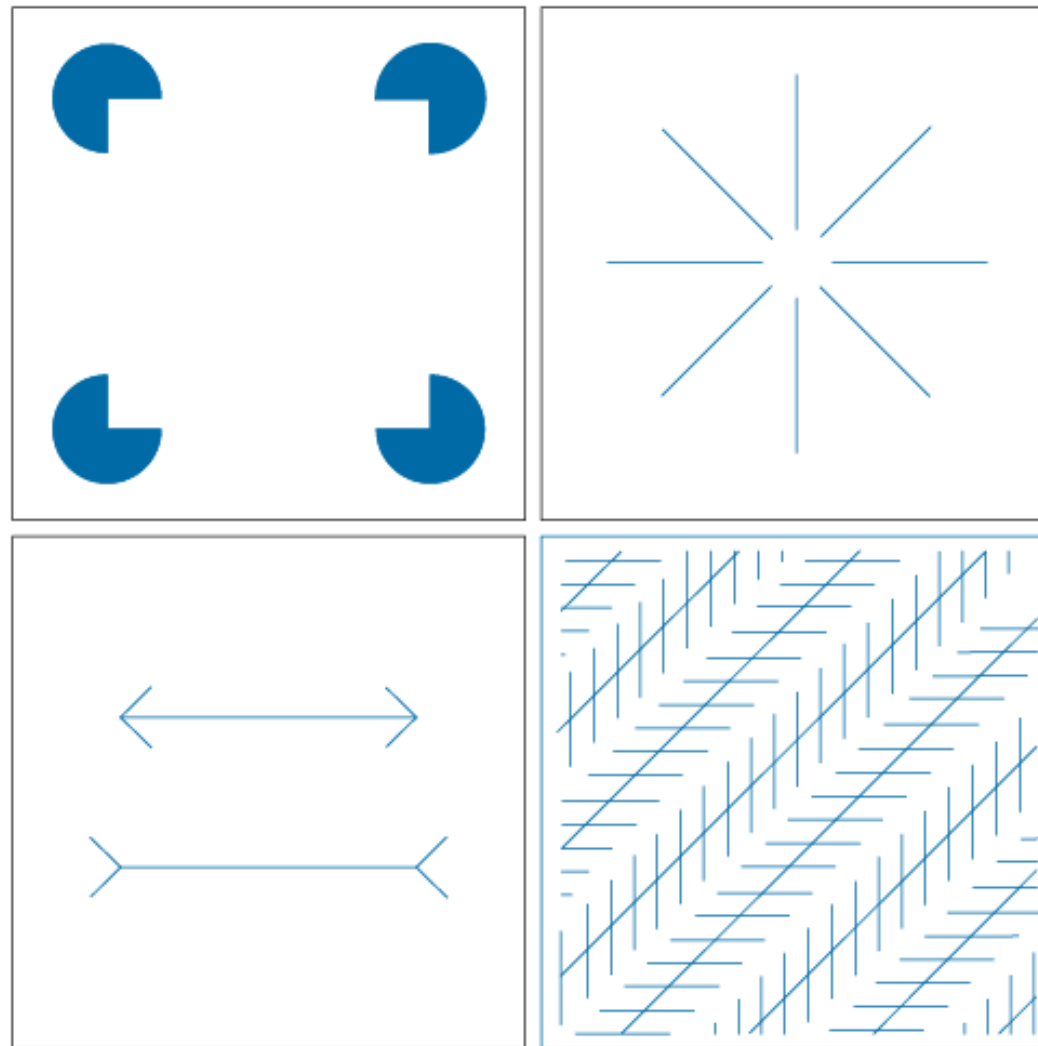


Example of simultaneous contrast

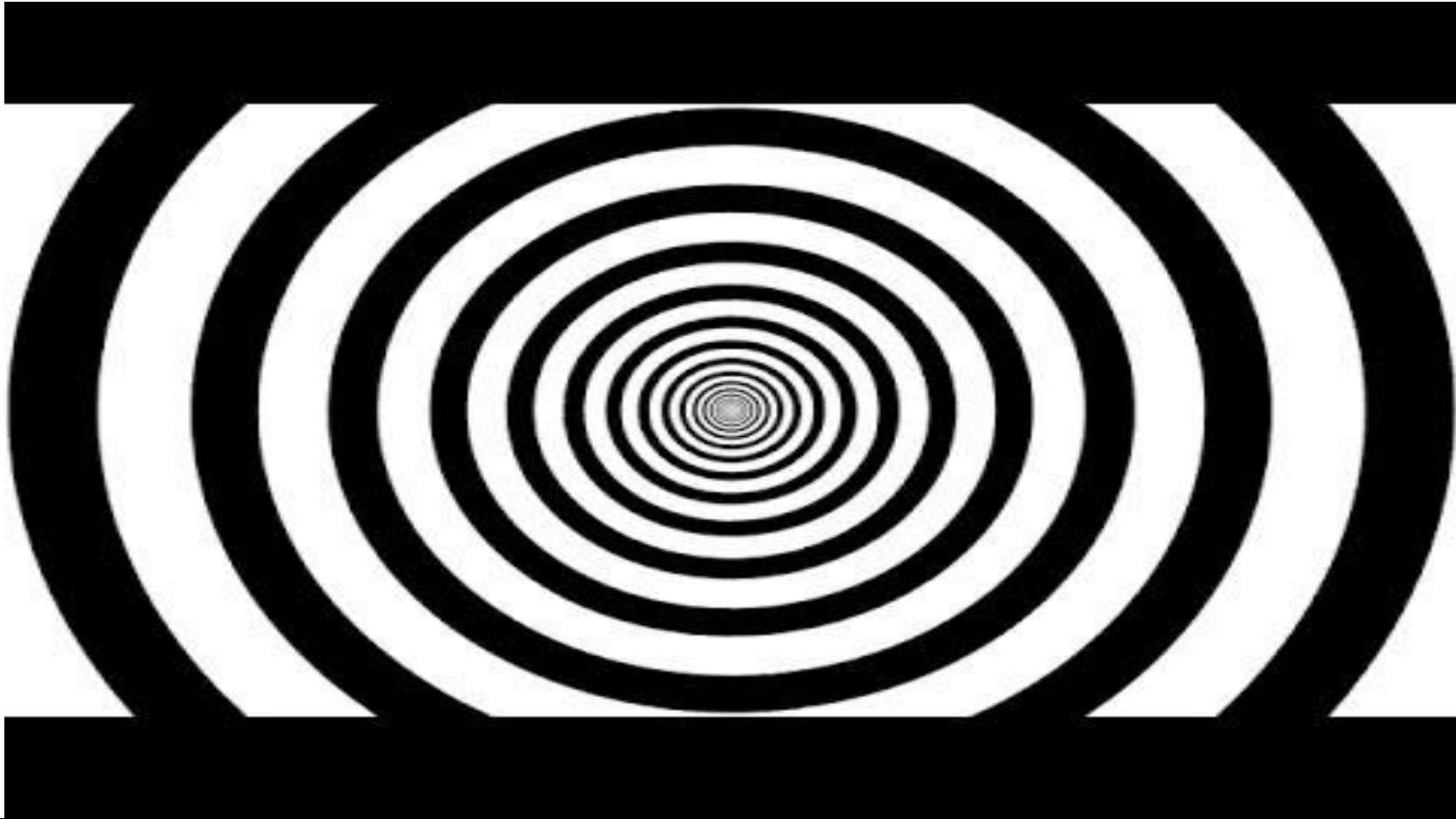


a b
c d

FIGURE 2.9 Some well-known optical illusions.



More Optical Illusions



Digital Images

Definition

- ❑ Picture

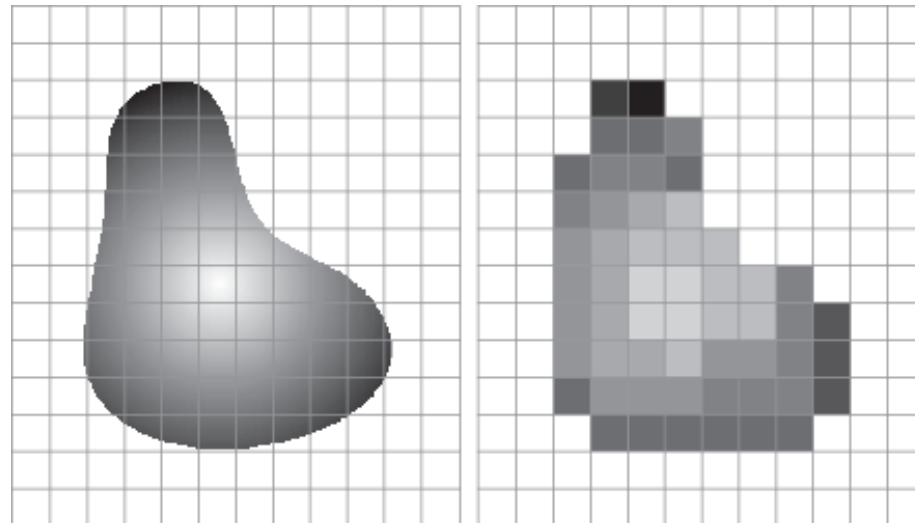
- ❑ A picture with its color (intensity) changes continuously

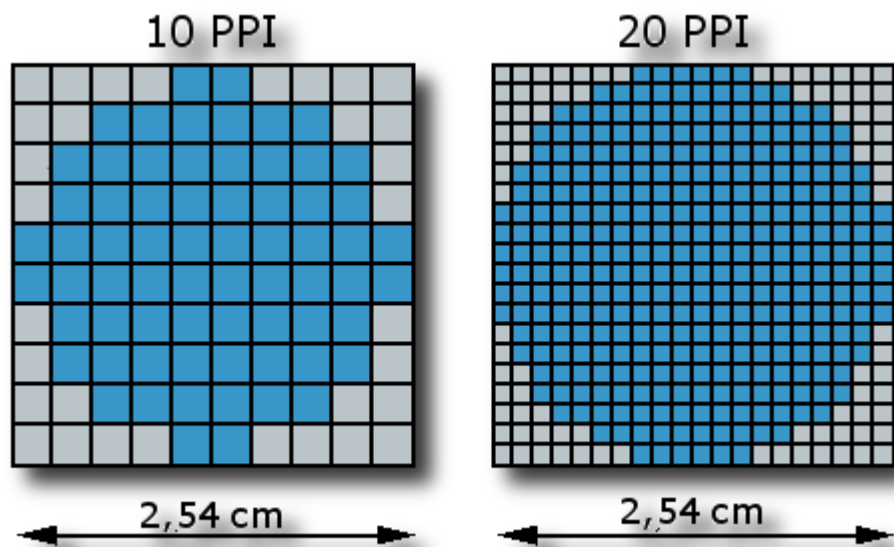
- ❑ Image

- ❑ An image can be obtained by digitalizing a picture, where the continuous values are sampled spatially and quantized into different intensity levels.

a b

FIGURE 2.17
(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.





dpi: dots per inch. 每英寸的墨點數量
用來描述印表機列印解析度

ppi : pixels per inch

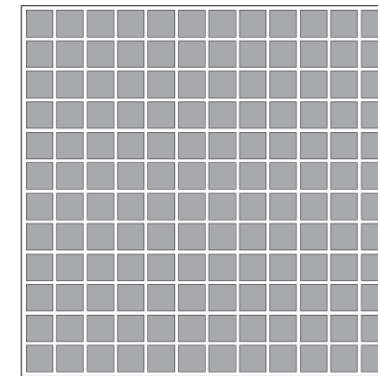
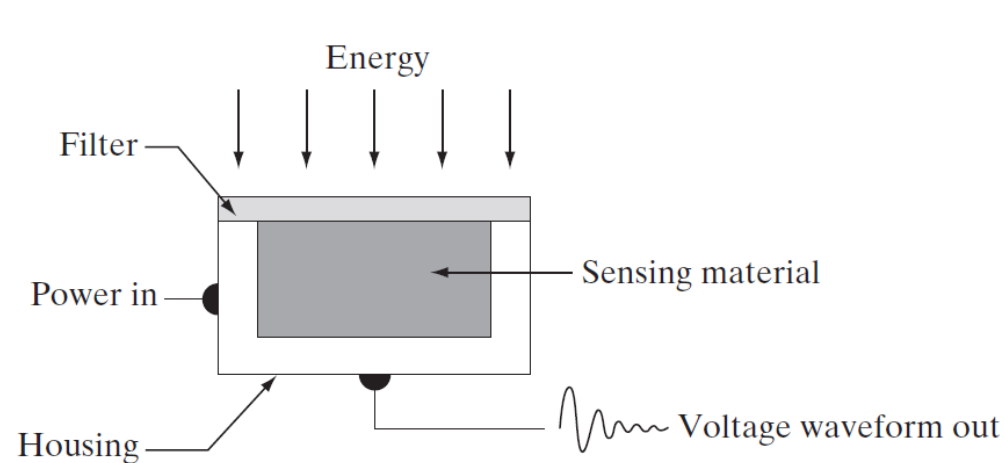
a	b
c	d

FIGURE 2.23
Effects of reducing spatial resolution. The images shown are at:
(a) 930 dpi,
(b) 300 dpi,
(c) 150 dpi, and
(d) 72 dpi.



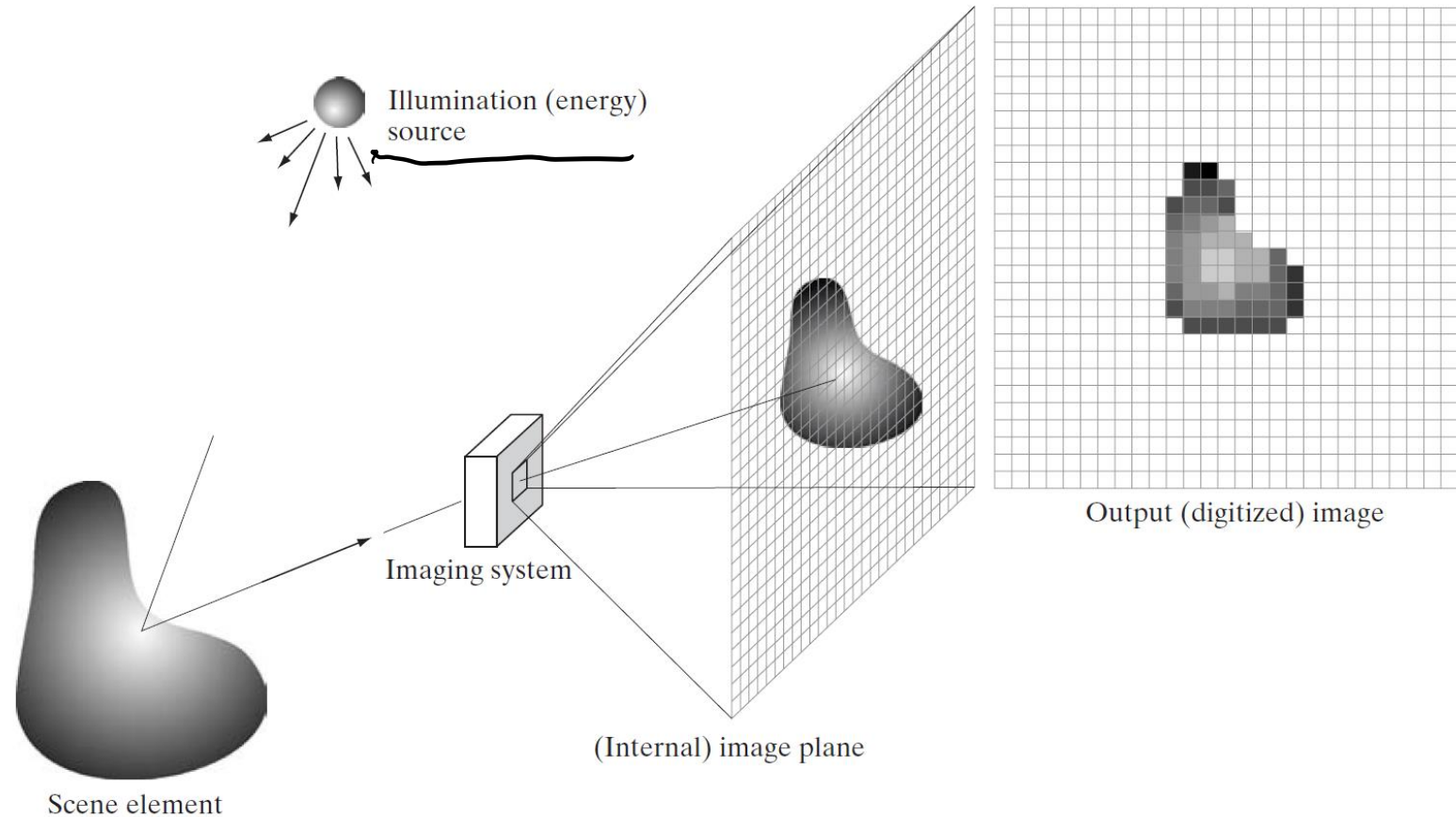
Where do digital images come from?

Through the interaction of an **illumination source** and the **reflection or absorption** of its energy by elements in the scene being imaged.



CCD sensors
(Charge-Coupled Device Sensors)

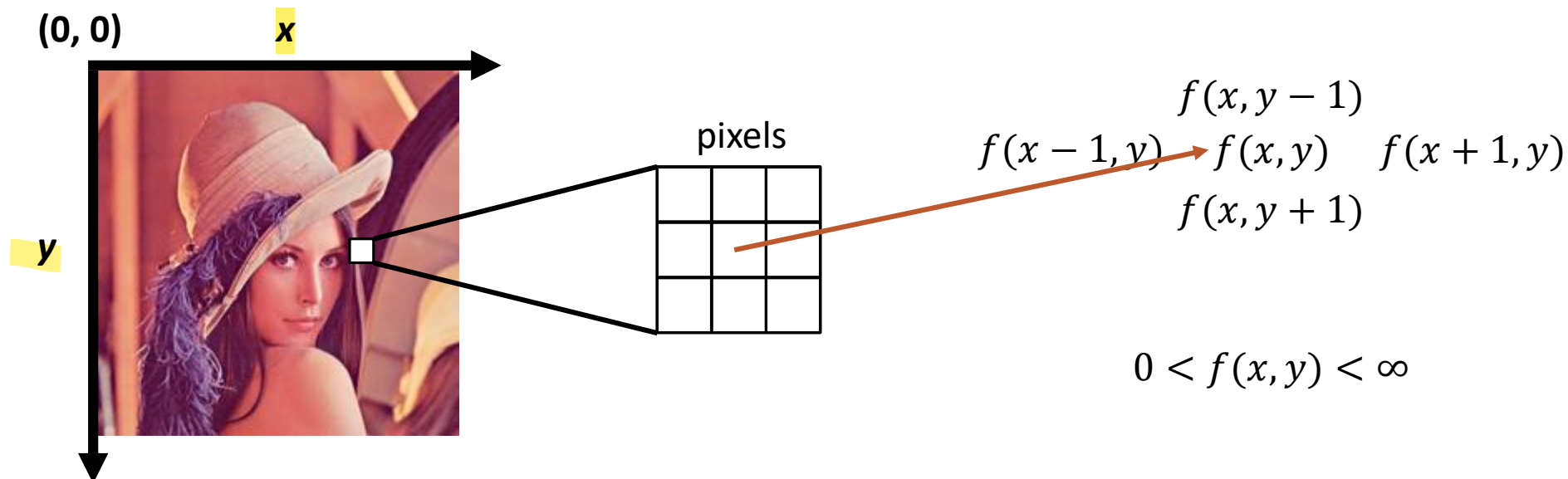
Simple Image Formation Model





x 軸向右延伸，y 軸向下延伸

Basic Relationships between Pixels



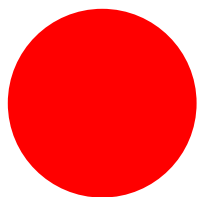
$f(x, y)$ may be characterized by (1) illumination and (2) reflectance.

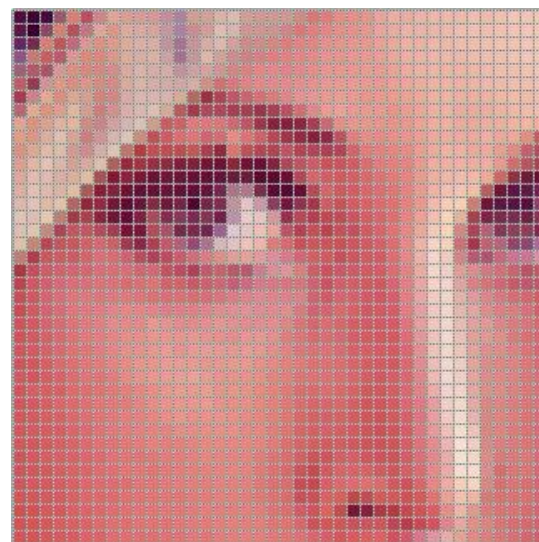
$$0 < i(x, y) < \infty \quad 0 < r(x, y) < 1$$

影像值 = 光照 × 反射率

$$f(x, y) = i(x, y)r(x, y)$$

Retinex model

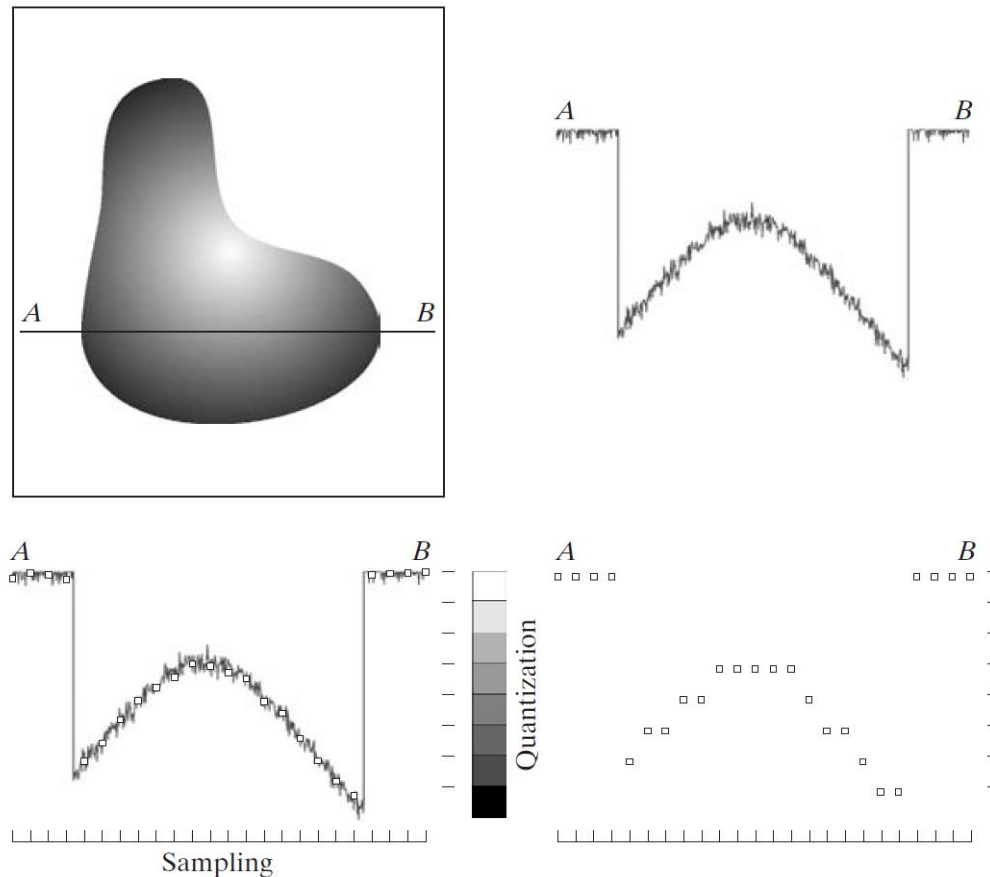




=



Image Sampling and Quantization



$$0 < f(x, y) < \infty$$



Sampling

$$f(x_0, y_0) = l$$



Quantization

$$L_{min} \leq l \leq L_{max}$$

Sampling & Aliasing in Image Processing

❑ Sampling: Converting Analog to Digital

- ❑ Sampling is the process of converting a **continuous (analog) signal** into a **discrete (digital) signal** by capturing values at regular intervals.
- ❑ **Higher sampling rates** retain more details, reducing loss of information.

❑ Aliasing: Distortion due to Undersampling

- ❑ **Occurs when the sampling rate is too low**, causing different signals to become indistinguishable (overlapping frequencies).
- ❑ **Results in visual artifacts**, such as jagged edges or moiré patterns in images.

❑ How to Prevent Aliasing?

- ❑ **Increase the Sampling Rate** → Follow **Nyquist Theorem** (sample at least **twice the highest frequency**).
- ❑ **Use Anti-Aliasing Filters** → Smooth out high-frequency components before sampling.

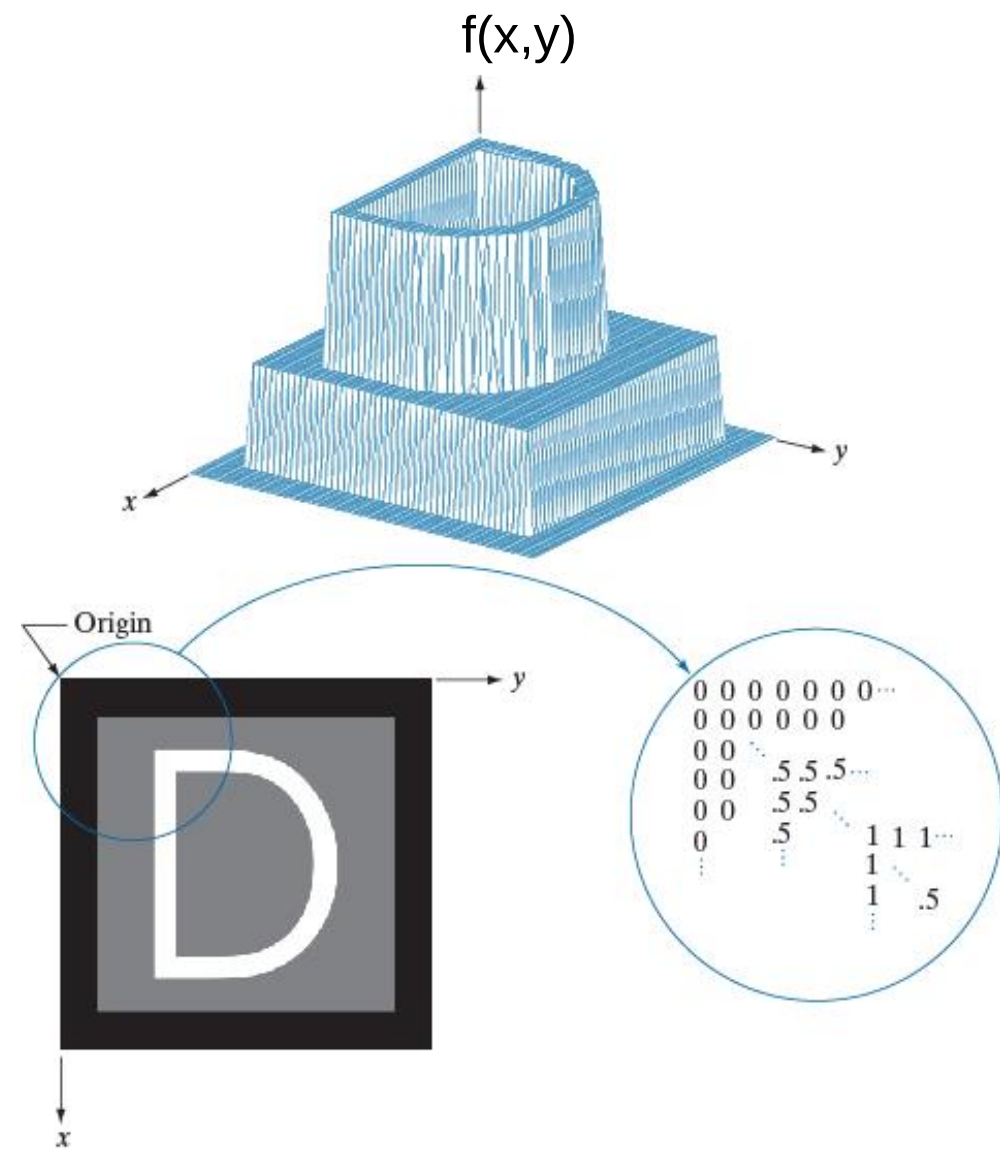
為什麼會混疊 (Aliasing) ?

混疊的原因是「高頻細節超過了取樣能夠解析的能力」。當你以太低的頻率對高頻圖像（如細條紋、鋸齒邊）取樣時，這些圖像資訊會被錯誤地重建為低頻圖樣，解法：使用抗混疊濾波器（Anti-Aliasing Filter），這是一種低通濾波器（Low-pass Filter）。原理：在取樣之前模糊掉（濾掉）圖像中超過取樣能力的高頻資訊

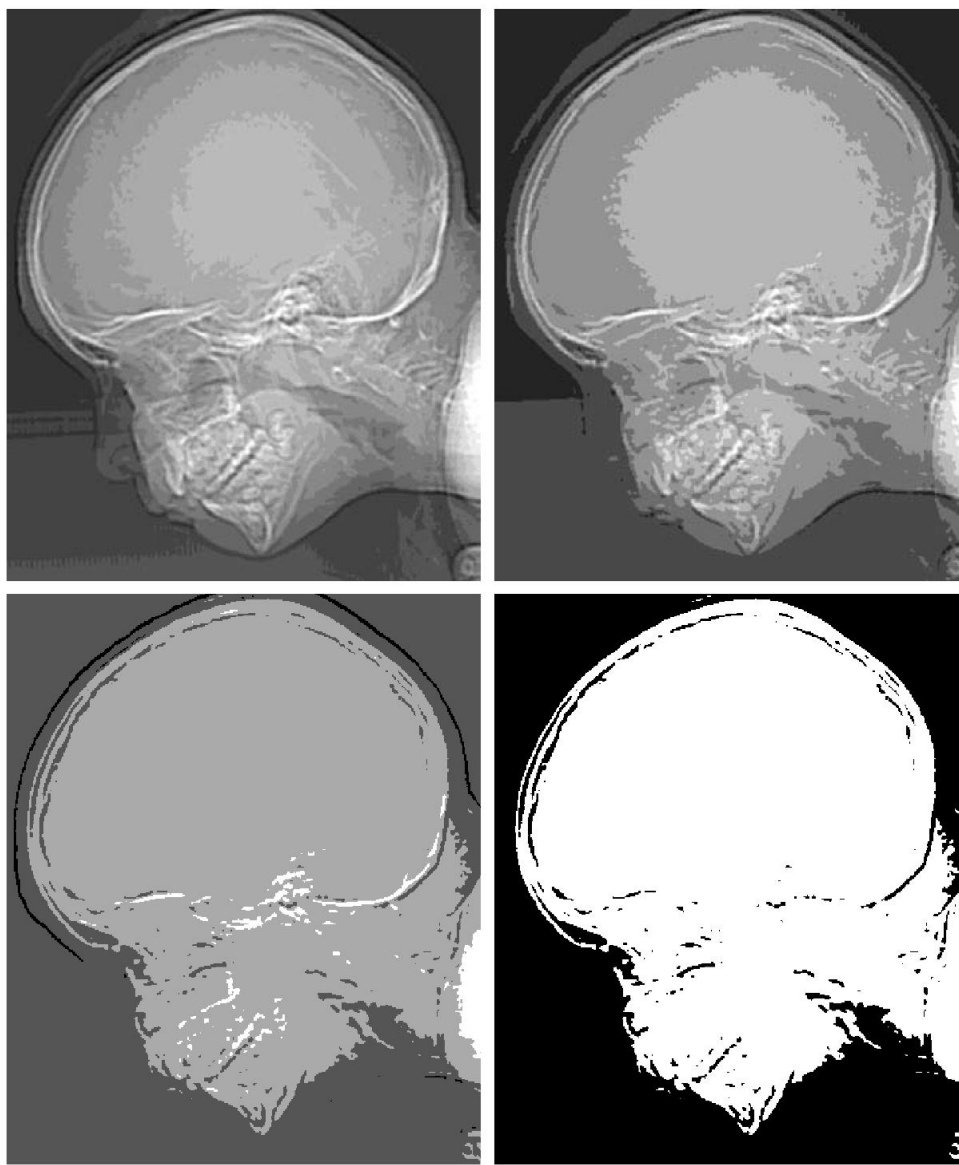
a
b c

FIGURE 2.18

(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)



three basic ways to represent $f(x,y)$.



e f
g h

FIGURE 2.21
(Continued)
(e)–(h) Image
displayed in 16, 8,
4, and 2 intensity
levels. (Original
courtesy of
Dr. David R.
Pickens,
Department of
Radiology &
Radiological
Sciences,
Vanderbilt
University
Medical Center.)

什麼是 Intensity Level (亮度等級) ?
在數位影像中，每一個像素都代表某個位置的亮度 (灰階) 或顏色。
而 intensity level 就是指：

一張影像中，像素亮度值可以取的「數值種類 (等級) 數量」

intensity level = 2 代表只有 0(黑),1(白)

Image Histogram

□ Count of # pixels of each intensity value

影像直方圖是用來顯示每個亮度值（intensity level）出現了多少像素。

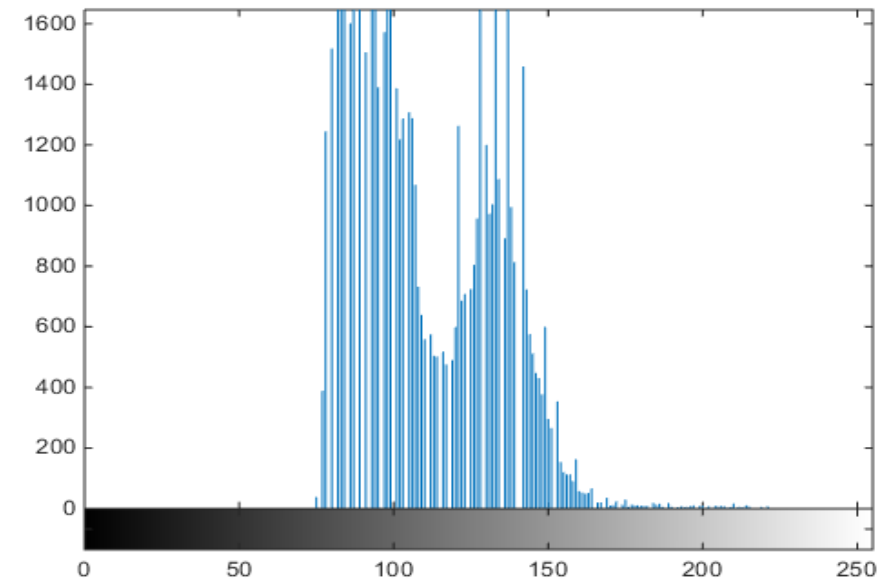


Histogram 偏左 => 圖像偏暗

Histogram 偏右 => 圖像偏亮

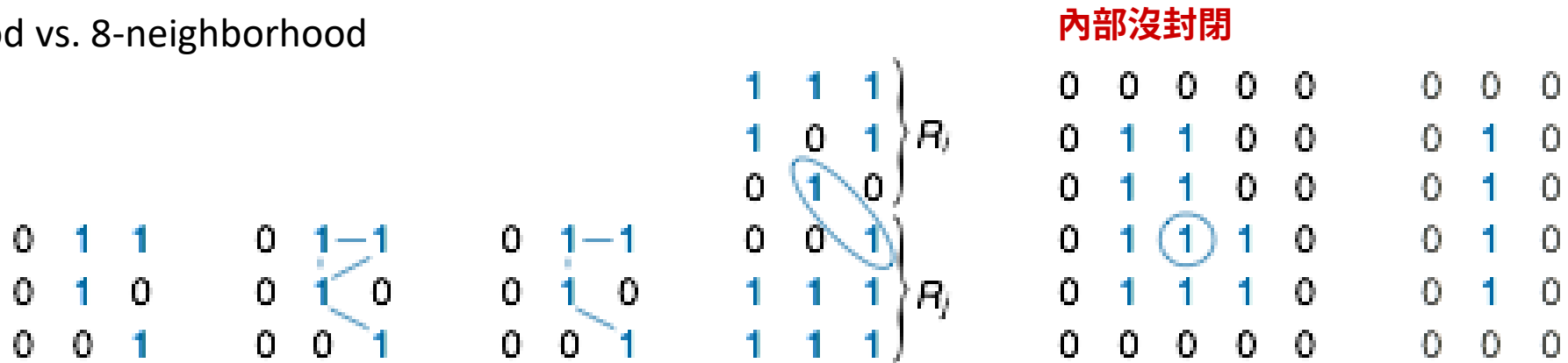
Histogram 集中中間、分佈狹窄 => 對比度低

Histogram 分佈廣泛 => 對比度好



Neighborhood Operations

4-neighborhood vs. 8-neighborhood



m-adjacency (混合鄰接)
一種避免 8-adjacency 中不必要連接的方式。
條件是：只當對角線之間沒有共通 4-鄰時才允許 8-adjacency。

a b c d e f

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

Distance Measures

For pixels p , q , and z , with coordinates (x, y) , (s, t) , and (v, w) , respectively, D is a *distance function* or *metric* if

- (a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$), 非負性 (Non-negativity)
- (b) $D(p, q) = D(q, p)$, and 對稱性 (Symmetry)
- (c) $D(p, z) \leq D(p, q) + D(q, z)$. 三角不等式 (Triangle Inequality)



Distance Measures

□ For pixels p , q , and z , with coordinates (x, y) , (s, t) , and (v, w) , respectively

□ Euclidean Distance between p and q

$$D_e(p, q) = \sqrt{[(x - s)^2 + (y - t)^2]} \quad \text{歐氏距離}$$

□ D_4 Distance: $D_4(p, q) = |x - s| + |y - t|$ 曼哈頓距離

□ D_8 Distance: $D_8(p, q) = \max(|x - s|, |y - t|)$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

D_4

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

D_8

(a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),

(b) $D(p, q) = D(q, p)$, and

(c) $D(p, z) \leq D(p, q) + D(q, z)$.

Mathematical Tools

Scalars, Vectors, and Matrices

- Scalars

- A scalar is just a number

- Vectors

- A vector is an array of numbers arranged in order.

$$x = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$$

Think of a vector as a point in the N dimensional space

- Matrices

- A matrix is a 2-D array of numbers

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Vector Spaces and Subspaces

- ❑ R^n : the set of all n-dimensional vectors (R: all components are real numbers)
- ❑ Vector space V
 - ❑ V is a set
 - ❑ if $u, v \in V$, then $u + v \in V$
 - ❑ if $u \in V$ and c is a scalar, $cu \in V$
- ❑ Subspace S of V
 - ❑ S is a subset of V
 - ❑ S is a vector space
- ❑ All the linear combinations of vectors in V must be in V
- ❑ Subspace S and U of V
 - ❑ $S \cap U$ is a subspace
 - ❑ $S \cup U$ may not be a subspace

Span

- Let u and v are vectors in S
 - $\text{span}\{u, v\}$: the set that has all linear combinations of u and v
- $\text{span}\{u, v\} = c_1 u + c_2 v, \forall c_1, c_2$

Ex:

$$\text{span}\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}\right\} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \forall c_1, c_2$$

$$\text{so, } \begin{bmatrix} 5 \\ 8 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \in \text{span}\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}\right\}$$

Norm

□ Given a vector space V , a norm on V is a function $f: V \rightarrow R$, such as

$\forall x, y \in V$

1. *Triangle inequality*: $f(x + y) \leq f(x) + f(y)$
2. *Absolute homogeneity*: $f(cx) = |c|f(x)$, for c is a scalar
3. *Positive definiteness*: if $f(x) = 0 \rightarrow x = 0$
4. *Non-negativity*: $f(x) \geq 0$

Different norms:

1. *L1 norm (Taxicab norm)*: $\|x\|_1 := \sum_{\forall i} |x_i|$
2. *L2 norm (Euclidean norm)*: $\|x\|_2 := \left(\sum_{\forall i} x_i^2 \right)^{\frac{1}{2}}$
3. *infinity norm*: $\|x\|_\infty := \max_i |x_i|$



Distance Measures

□ For pixels p, q , and z , with coordinates (x, y) , (s, t) , and (v, w) , respectively

□ **Euclidean Distance** between p and q

$$D_e(p, q) = L_2(p - q) = [(x - s)^2 + (y - t)^2]^{\frac{1}{2}}$$

□ D_4 Distance: $D_4(p, q) = L_1(p - q) = |x - s| + |y - t|$

□ D_8 Distance: $D_8(p, q) = L_\infty(p - q) = \max(|x - s|, |y - t|)$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

D_4

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

D_8

(a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),

(b) $D(p, q) = D(q, p)$, and

(c) $D(p, z) \leq D(p, q) + D(q, z)$.



Array/Matrix Operations

☐ Array Product Hadamard Product

☐ Pixel-by-pixel basis $\odot$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

常用在影像處理中的：

遮罩應用 (masking)

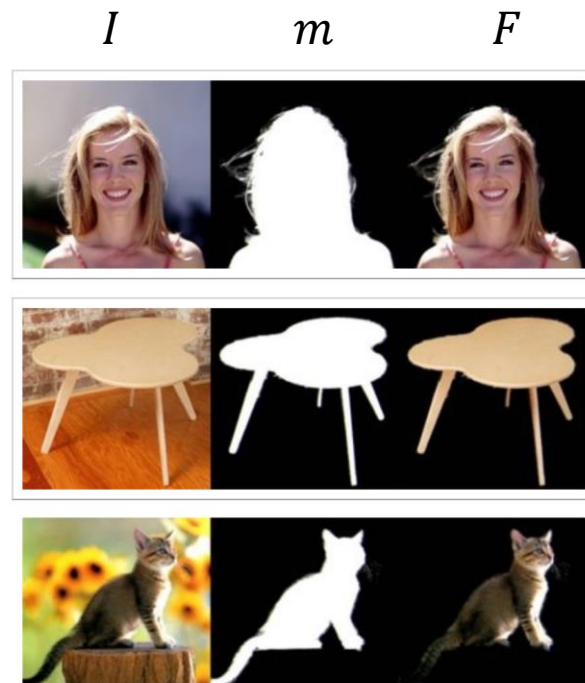
權重乘法 (weighted filters)

☐ Matrix Product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Array Operations

□ Example $I, m, F \in R^{h \times w}$



$$F = I \odot m$$

$\odot$: element-wise multiplication

Image Matting (array product)

Linear Operations

- ❑ Consider a general operator: H

$$H[f(x, y)] = g(x, y)$$

- ❑ H : linear

- ❑ satisfies the principles of **linearity**:

- ❑ **Additivity**: The response to the sum of two inputs equals the sum of their individual responses.

- ❑ **Homogeneity (Scaling)**: Scaling the input scales the output by the same factor.

$$H[a_i f_i(x, y) + a_j f_j(x, y)] = a_i H[f_i(x, y)] + a_j H[f_j(x, y)]$$

$$f_z = a_i g_i(x, y) + a_j g_j(x, y)$$

$$g_z$$

- ❑ Example: $\sum_x (3f(x) + 4g(x))$

Nonlinear Operations

□ H: nonlinear

$$H[a_i f_i(x, y) + a_j f_j(x, y)] \neq a_i H[f_i(x, y)] + a_j H[f_j(x, y)]$$

□ Example:

$$f_1 = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad f_2 = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix}$$

$$\max(f_1 - f_2) \neq \max(f_1) - \max(f_2)$$

s&p noise 會是極端值0,255所以用均值濾波效果不好，用中值(中位數)濾波更好

Linear versus Nonlinear Operations

Linear

- mean/average filters or Gaussian filtering



Original Image



Image with s&p
noise

Nonlinear

- median filtering



Mean filtering



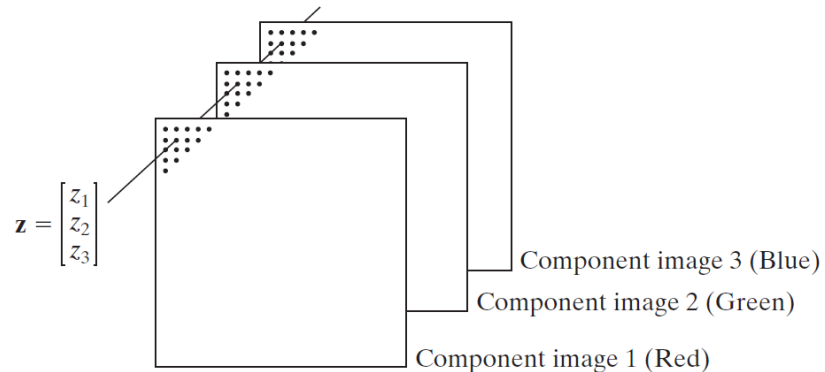
Median filtering

Vector & Matrix Operations

- Color images are mostly formed in RGB color spaces by red, blue, and green components:

$\mathbf{z}$ represents a pixel in a column vector

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$



A RGB color image of size $M \times N$ can be represented as: $I^{M \times N \times 3}$

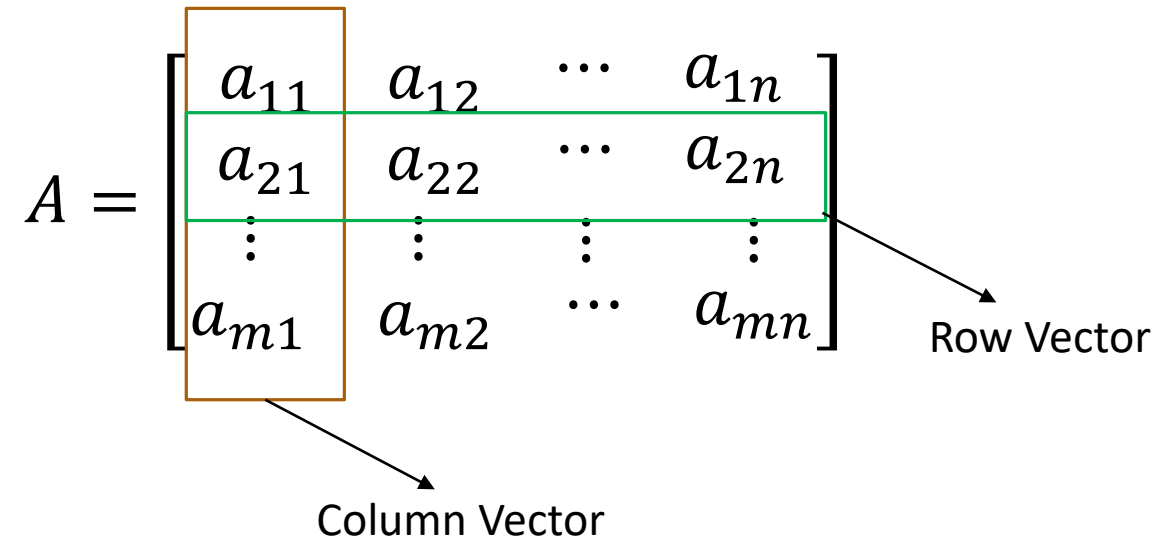
More Definitions

An $m \times n$ (read "m by n") **matrix**, denoted by $\mathbf{A}$, is a rectangular array of entries or elements (numbers, or symbols representing numbers) enclosed typically by square brackets, where m is the number of rows and n the number of columns.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Row Vector

Column Vector

The diagram shows a matrix A enclosed in large square brackets. Inside, the first row contains elements a_{11}, a_{12}, an ellipsis, and a_{1n}. The second row contains a_{21}, a_{22}, an ellipsis, and a_{2n}. Below these are three vertical dots, followed by the m-th row containing a_{m1}, a_{m2}, an ellipsis, and a_{mn}. A green rectangular box highlights the first two rows, with an arrow pointing to the label 'Row Vector'. An orange rectangular box highlights the first two columns, with an arrow pointing to the label 'Column Vector'.

- **Square matrix $\mathbf{A}$:** $m = n$.
- **Diagonal matrix $\mathbf{A}$?**
- **Identity matrix $\mathbf{A}$** is also a diagonal matrix.
- **Null/Zero matrix $\mathbf{A}$:** $\mathbf{A} = \mathbf{0}$
- Trace of $\mathbf{A}$: $Tr(\mathbf{A}) = \sum_{i=1}^n a_{ii}$
- $\mathbf{A} = \mathbf{B}$: $a_{ij} = b_{ij}$, $\forall i, j$
- $\mathbf{A}^T$: the transpose of $\mathbf{A}$. $(\mathbf{A})_{i,j}^T = A_{j,i}$
- $\mathbf{A}^T = \mathbf{A}$: $\mathbf{A}$ is symmetric.
- $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ and $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$: $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$.

Properties of the Trace of a Matrix

- ❑ Symmetry Property

- ❑ $Tr(A) = Tr(A^T)$

- ❑ Scalar Multiplication

- ❑ $Tr(kA) = kTr(A)$

- ❑ Cyclic Property

- ❑ $Tr(AB) = Tr(BA)$

- ❑ Additivity

- ❑ $Tr(A + B) = Tr(A) + Tr(B)$

佛羅貝尼烏斯範數

Frobenius Norm

是一種用來衡量"整個矩陣大小（或能量）"的方式。
可以把它想成是「矩陣版本的歐氏範數（L2 norm）」。

□ The Frobenius Norm (Euclidean Norm)

□ Definition:

Let $A \in R^{m \times n}$

把所有元素平方加總後開根號，代表整體大小或誤差。

$$\|A\|_F \equiv \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{i,j}|^2} = \sqrt{\text{Tr}(AA^H)}$$

A^H : the conjugate transpose of A

共軛轉置

有什麼用？

1. 比較矩陣大小

2. 圖像誤差計算 $\Rightarrow \|A-B\|_F$ (量化整張圖有多不一樣)

Set Operations

We denote the fact that an *element* a *belongs* to set A by

$$a \in A$$

If a is not an element of A , then we write

$$a \notin A$$

A set can be specified by listing all of its elements, or by listing properties common to all elements. For example, suppose that I is the set of all integers. A set B consisting the first five nonzero integers is specified using the notation

$$B = \{1, 2, 3, 4, 5\}$$

The set of all integers less than 10 is specified using the notation

$$C = \{c \in I \mid c < 10\}$$

which we read as "C is the set of integers such that each members of the set is less than 10." The "such that" condition is denoted by the symbol " $\mid$ ". As shown in the previous two equations, the elements of the set are enclosed by curly brackets.

The set with no elements is called the *empty* or *null set*, denoted in this review by the symbol $\emptyset$.

Two sets A and B are said to be *equal* if and only if they contain the same elements. Set equality is denoted by

$$A = B$$

If the elements of two sets are not the same, we say that the sets are *not equal*, and denote this by

$$A \neq B$$

If every element of B is also an element of A , we say that B is a *subset* of A :

$$B \subseteq A$$

Universal set, denoted by U , a set to contain all the elements

$$A \subseteq U$$

The operations on sets associated with basic probability theory are straightforward. The *union* of two sets A and B , denoted by

$$A \cup B$$

is the set of elements that are either in A or in B , or in both. In other words,

$$A \cup B = \{z \mid z \in A \text{ or } z \in B\}$$

Similarly, the *intersection* of sets A and B , denoted by

$$A \cap B$$

is the set of elements common to both A and B ; that is,

$$A \cap B = \{z \mid z \in A \text{ and } z \in B\}$$

Two sets having no elements in common are said to be *disjoint* or *mutually exclusive*, in which case

$$A \cap B = \emptyset$$

The *complement* of set A is defined as

$$A^c = \{z \mid z \notin A\}$$

Clearly, $(A^c)^c = A$. Sometimes the complement of A is denoted as $\overline{A}$.

The *difference* of two sets A and B , denoted $A - B$, is the set of elements that belong to A , but not to B . In other words,

$$A - B = \{z \mid z \in A, z \notin B\}$$

Some Important Set Relationships

$$S^c = \emptyset; \emptyset^c = S;$$

$$A \cup A^c = S; A \cap A^c = \emptyset$$

$$A \cup \emptyset = A; A \cap \emptyset = \emptyset; S \cup \emptyset = S; S \cap \emptyset = \emptyset$$

$$A \cup A = A; A \cap A = A; A \cup S = S; A \cap S = A$$

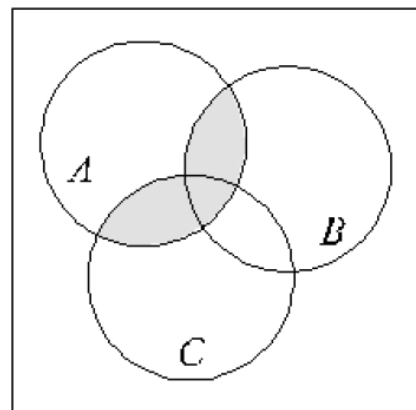
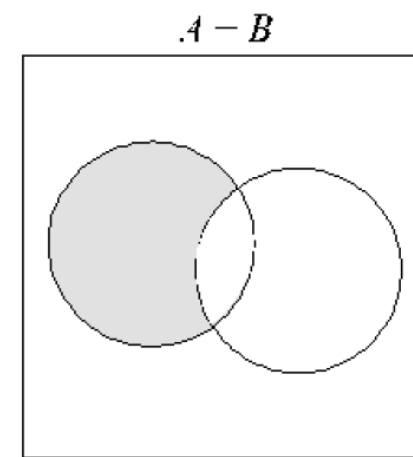
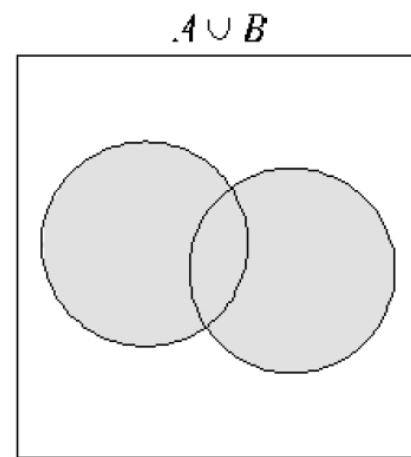
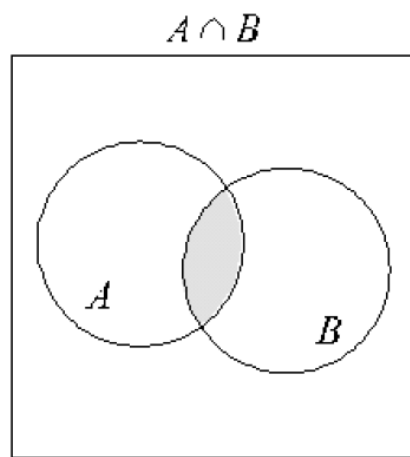
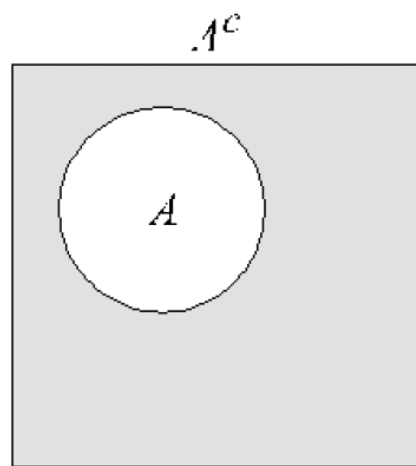
$$A \cup B = B \cup A; A \cap B = B \cap A$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

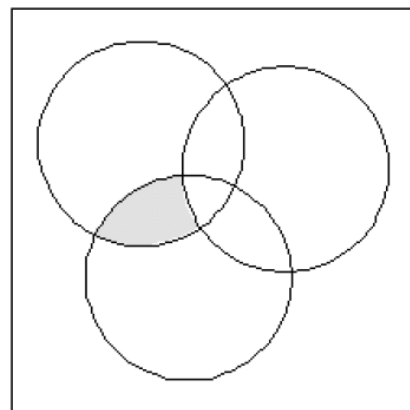
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(A \cup B) \cup C = A \cup (B \cup C) = A \cup B \cup C$$

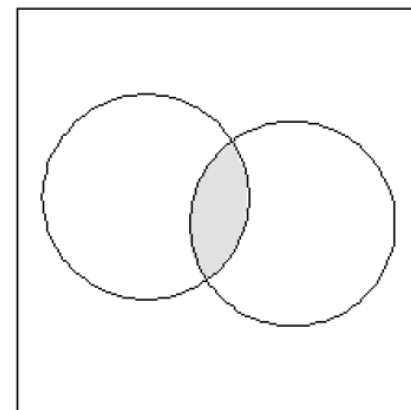
$$(A \cap B) \cap C = A \cap (B \cap C) = A \cap B \cap C$$



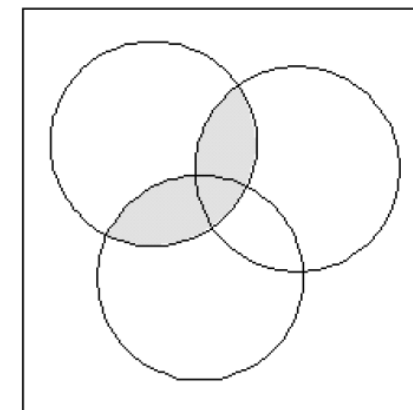
$A \cap (B \cup C)$



$(A \cap C) - (A \cap B \cap C)$



$A \cap B$



$(A \cap B) \cup (A \cap C) - (A \cap B \cap C)$

Set Operations

□ A: a set composed of elements; a: an element that belongs to A $\rightarrow a \in A$

A, B, C, and D are all sets:

$$A \subseteq B$$

subset

$$C = A \cup B$$

union

$$D = A \cap B$$

intersection

$$A \cap B = \emptyset$$

disjoint/mutual exclusive

The complement of a set A

$$A^c = \{w | w \notin A\}$$

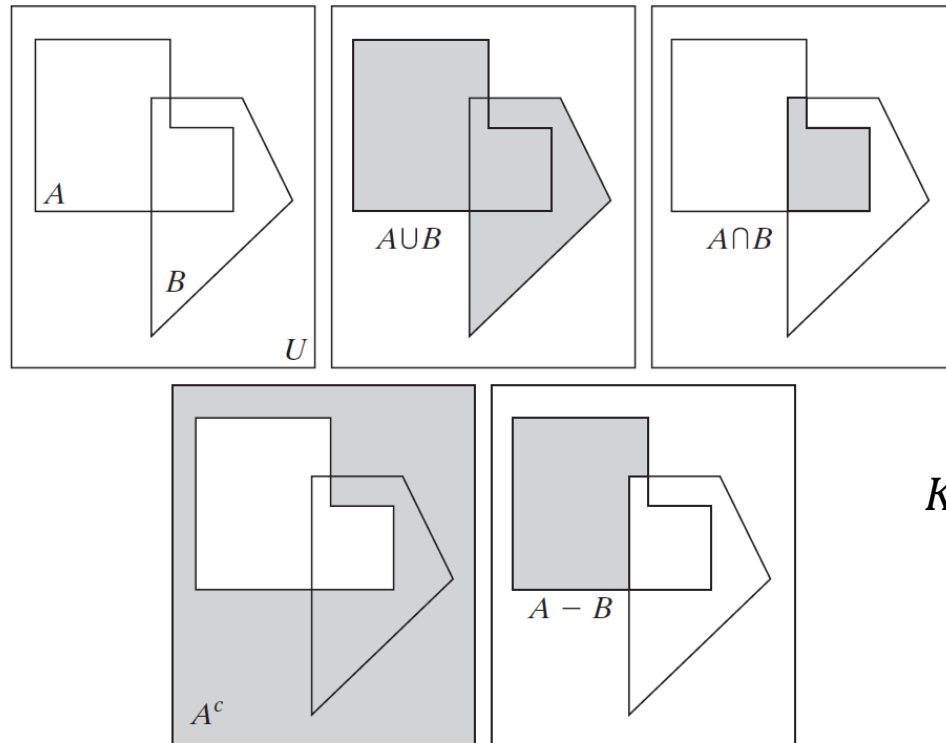
or $(\bar{A})$

The difference of two set A and B

$$A - B = \{w | w \in A, w \notin B\} = A \cap B^c$$

$$A^c = U - A$$

Set Operations

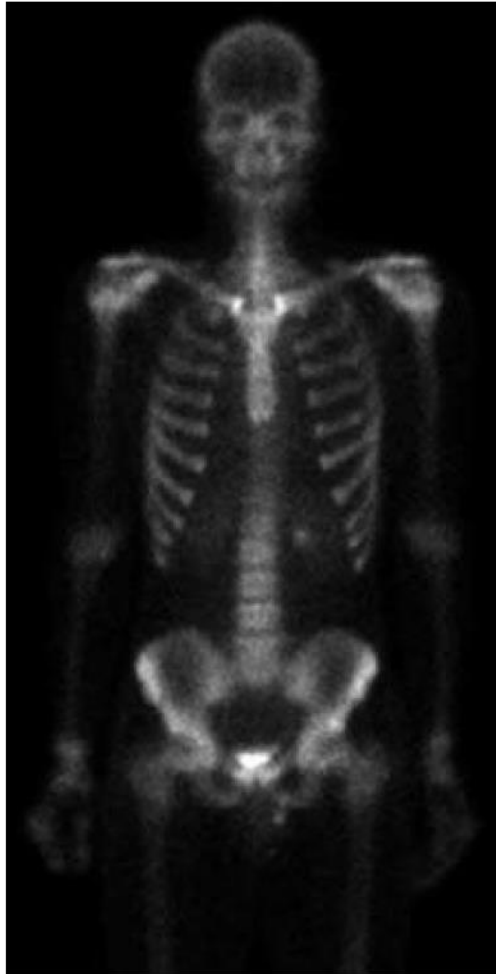


Let the elements of a **gray-scale** image be represented by a set A whose elements are triplets of the form (x, y, z) , where x and y are spatial coordinates and z denotes intensity.

$$A^c = \{(x, y, K - z) | (x, y, z) \in A\}$$

($\bar{A}$)

$K = 2^k - 1$, where k is the number of intensity bits used to represent z



A



$\bar{A}$

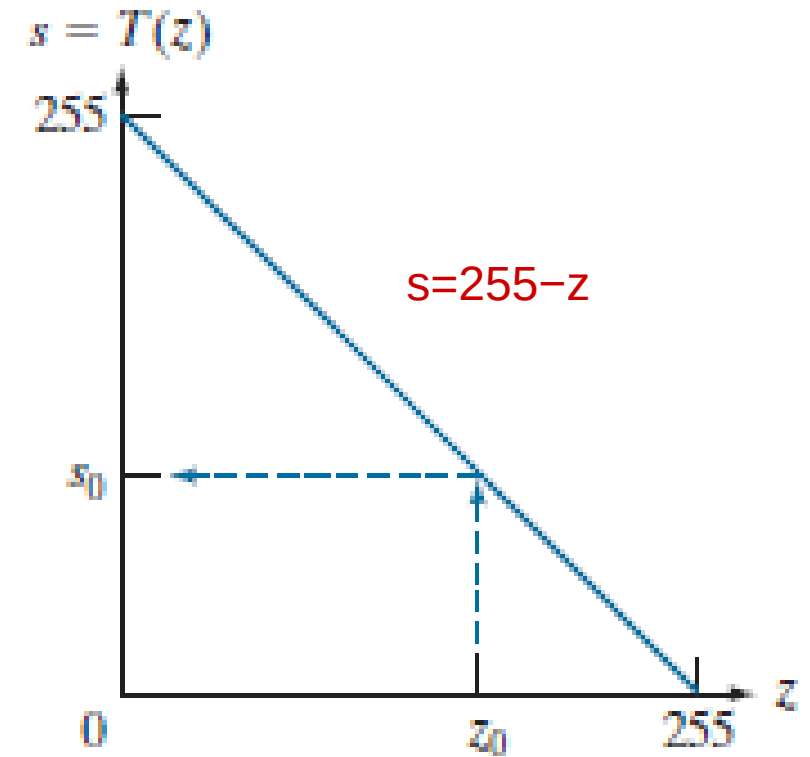


0

$2^k - 1$

影像負片 (photographic negative)

FIGURE 2.38
Intensity
transformation
function used to
obtain the digital
equivalent of
photographic
negative of an
8-bit image..



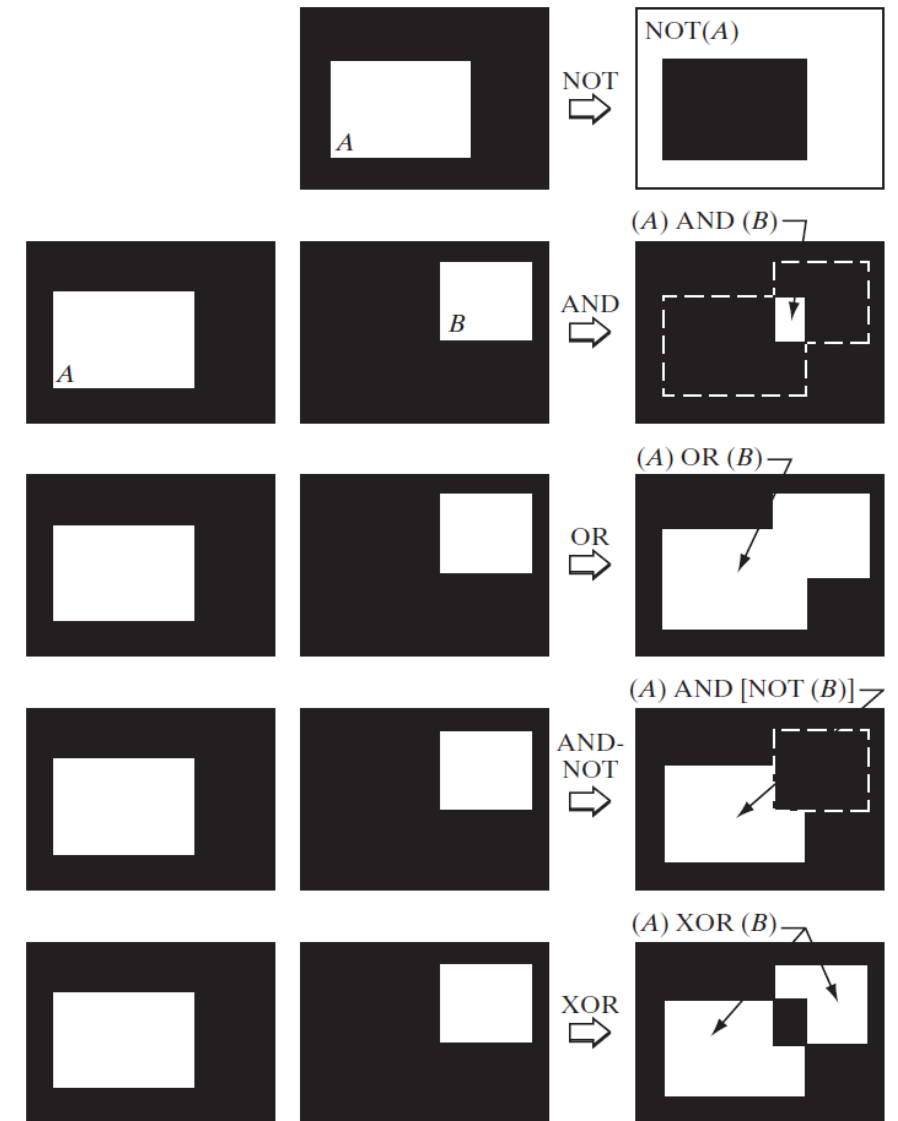
Logical Operations on Binary Images

TABLE 2.2

Truth table defining the logical operators AND($\wedge$), OR($\vee$), and NOT($\sim$).

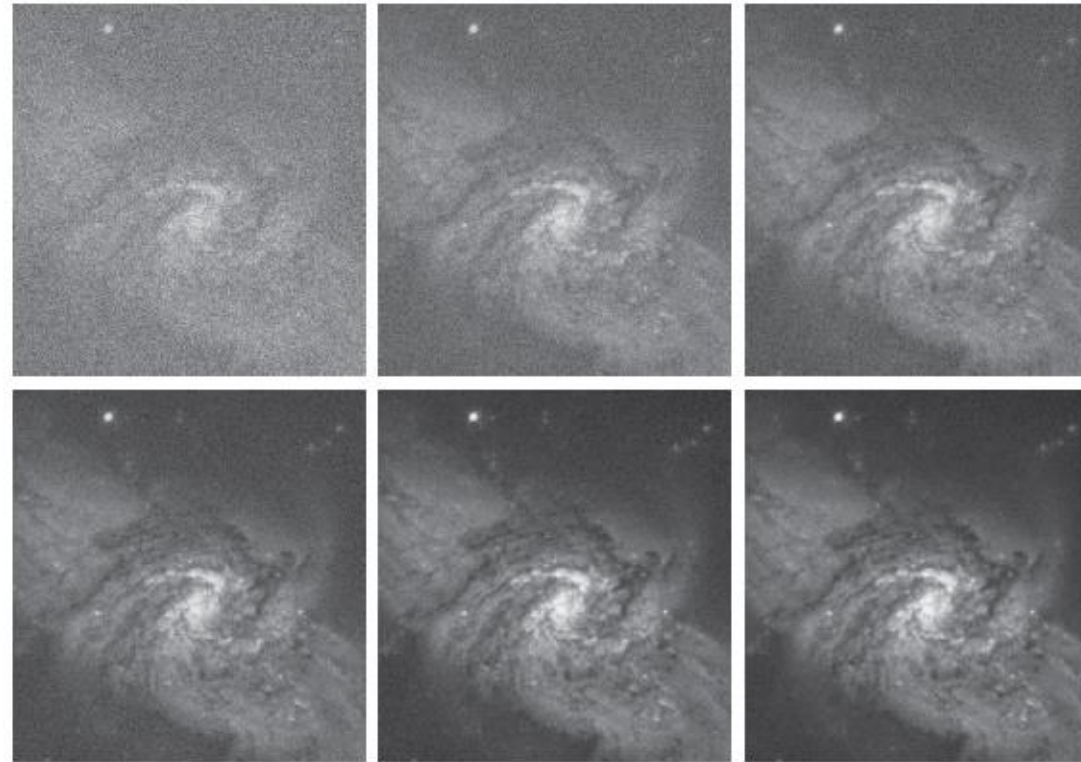
a	b	$a \text{ AND } b$	$a \text{ OR } b$	$\text{NOT}(a)$
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

$A \text{ xor } B = 1$ if $(A = 1 \text{ or } B = 1)$ and $(AB \neq 1)$



可以用來消除高斯雜訊

Pixel Addition and Normalization



所謂的高斯雜訊，就是你將雜訊轉換到frequency domain 來看，它分佈的狀態是會呈現所謂的高斯分佈(Gaussian distribution)，也稱做常態分佈

a b c
d e f

FIGURE 2.29 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566×598 pixels, and all were scaled so that their intensities would span the full $[0, 255]$ intensity scale. (Original image courtesy of NASA.)

Region of interest

ROI Segmentation

Array Product



a b c

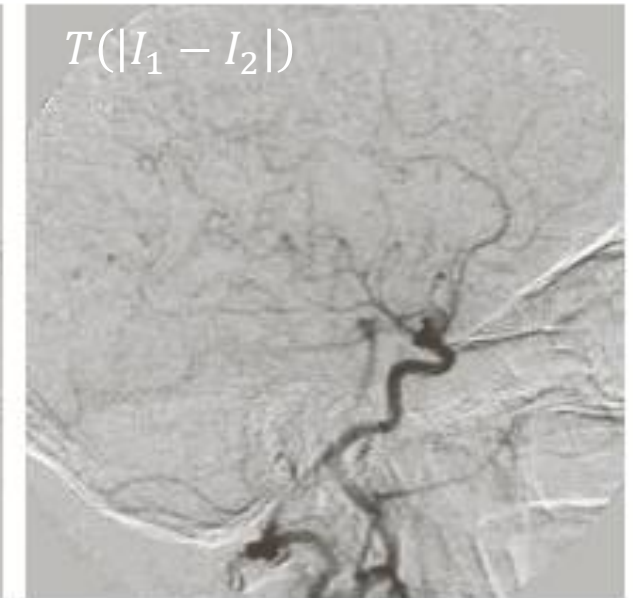
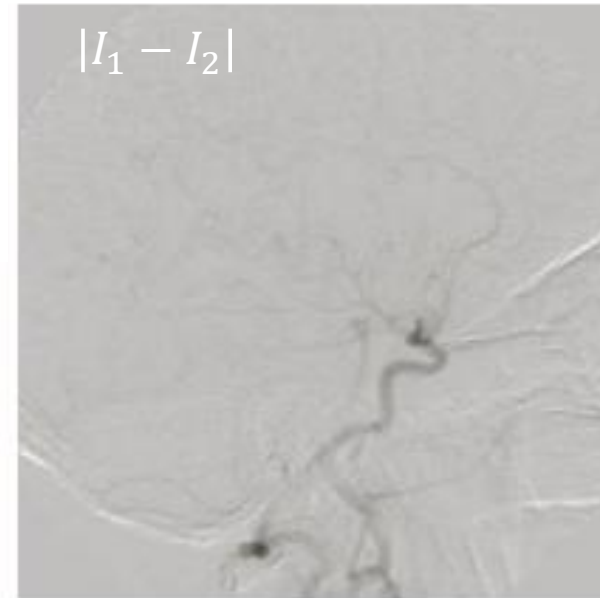
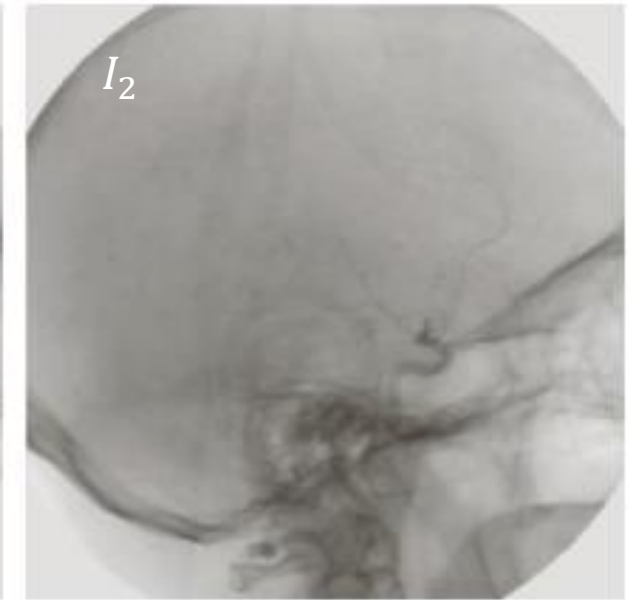
FIGURE 2.34 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

Pixel Difference

I_1 : 遮罩影像 (mask image, 無造影劑)
 I_2 : 實況影像 (live image, 加了造影劑)
 $|I_1 - I_2|$: 兩影像差異 (像素級絕對差)
 $T(|I_1 - I_2|)$: 差異影像經過強化 (轉換增強細節)

a	b
c	d

FIGURE 2.32
Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of the Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)

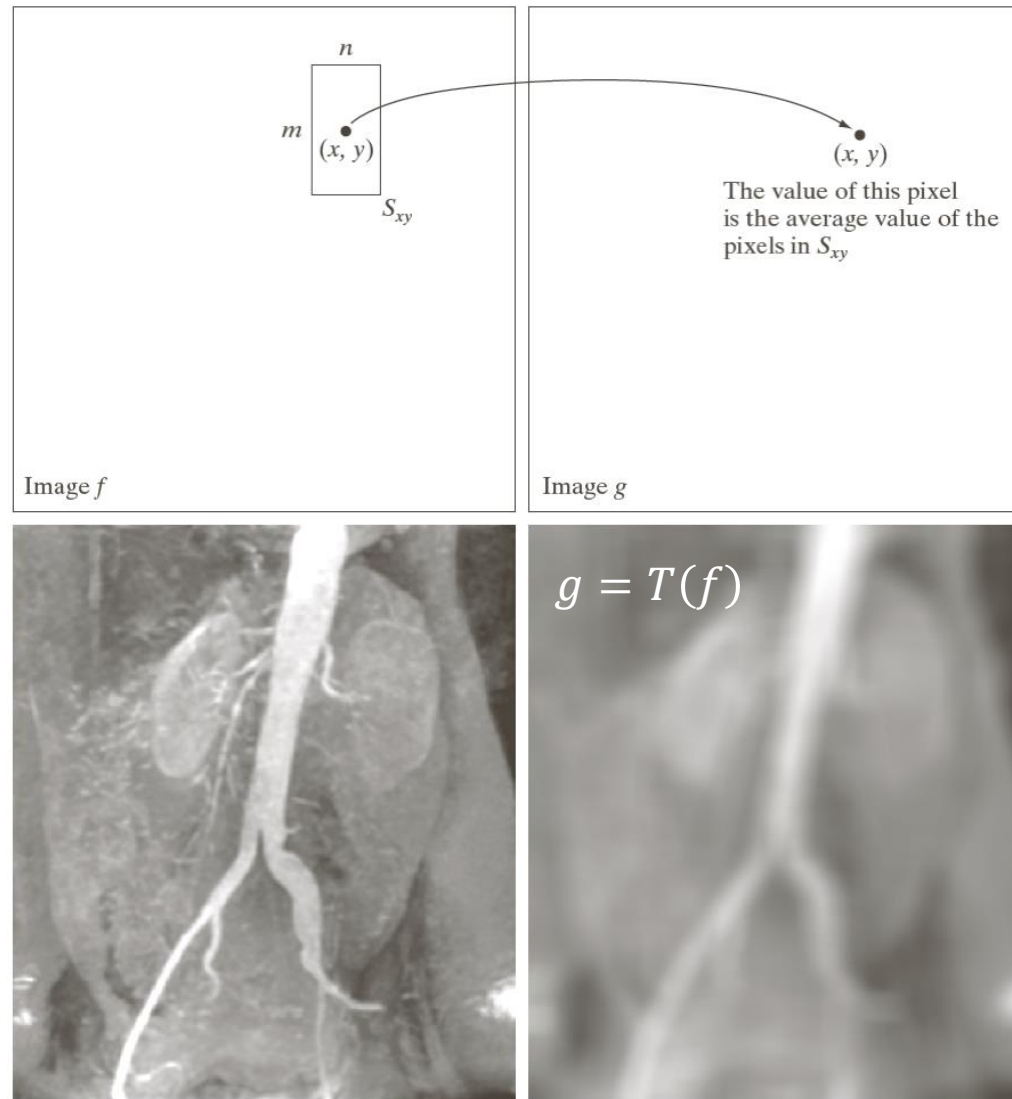


T : Transformation function
for enhancement

a b
c d

FIGURE 2.39

Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (see Section 1.3). (d) The result of using Eq. (2-43) with $m = n = 41$. The images are of size 790×686 pixels. (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)



Let Ω_{xy} denote the set of coordinates of a neighborhood centered on an point (x, y) in an image f

$$g(x, y) = T(f(x, y)) = \frac{\sum_{(r,c) \in \Omega_{xy}} f(r, c)}{mn}$$

Miniature Effect

- ❑ Capturing a scene from a **high angle** combined with **boundary blur effects** creates the **illusion of a miniature model**, a technique known as **Tilt-Shift Photography**.
 - ❑ **High-Angle Perspective** → Mimics the way we look at small-scale models.
 - ❑ **Selective Focus & Boundary Blur** → Simulates a **shallow depth of field**, making the scene appear tiny.
 - ❑ **Increased Contrast & Saturation** → Enhances the **toy-like effect** by making colors pop.
- ❑ **Human Perception:** Our eyes associate **shallow depth of field** with close-up views, tricking the brain into thinking the scene is small.



<https://www.smashingmagazine.com/2008/11/beautiful-examples-of-tilt-shift-photography/>

卷積是一種在影像處理中用來進行模糊、銳化、邊緣偵測等操作的基本工具。

Convolution

$$g(x, y) = T(f(x, y)) \rightarrow g = f \circ T$$

Stride = 1
Kernel = 3*3

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

f



1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Kernel



1/3	1/3	1/3	1/3
1/3	1/3	4/9	1/3
1/3	1/3	4/9	1/3
1/3	2/9	4/9	1/3

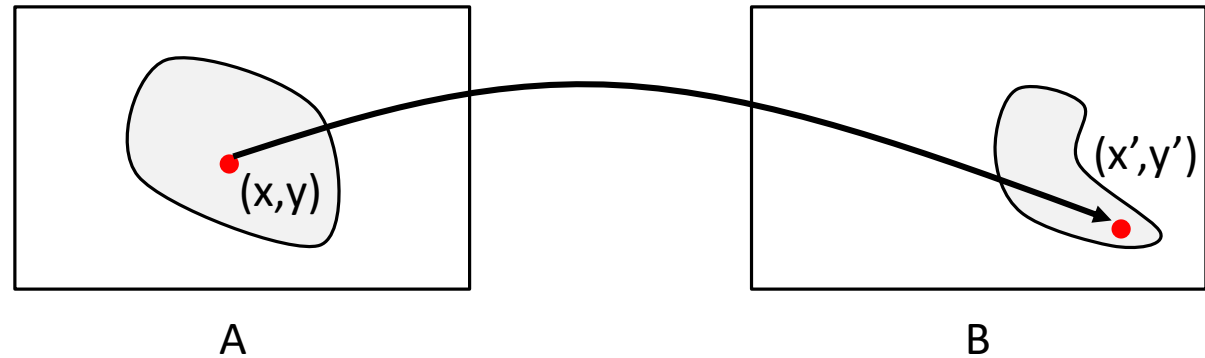
Result

convolved with T

Geometric Transformations on Images

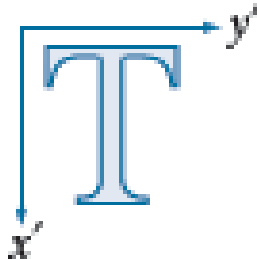
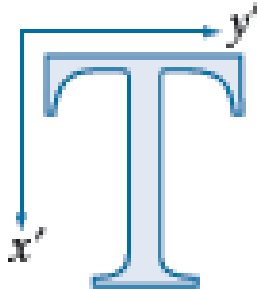
- Geometric Transformations
 - Spatial transformation of coordinates
 - Each point in the image A is mapped to another point in another image B

$$(x', y') = T((x, y))$$



Geometric Transformations on Images

Affine transformation $\begin{bmatrix} x' & y' & 1 \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} & t_x \\ t_{21} & t_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	

Scaling + Translation

整數座標情況

$$[x' \ y' \ 1] = \mathbf{T} \begin{bmatrix} 4 \\ 40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 & 0 & 10 \\ 0 & 0.25 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 40 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 11 \\ 20 \\ 1 \end{bmatrix}$$

The transformed pixel coordinate is integer

Intensity:

$$g(11, 20) = f(4, 40)$$

Coordinates: (4, 40) → (11, 20)

對應到整數像素座標 => 不需插值

Scaling + Translation

$$[x' \ y' \ 1] = \mathbf{T} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 & 0 & 10 \\ 0 & 0.25 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 10.25 \\ 10.25 \\ 1 \end{bmatrix}$$

Coordinates: (1, 1) $\rightarrow$ (10.25, 10.25)

會出現非整數座標問題

The transformed pixel coordinate is not integer!!

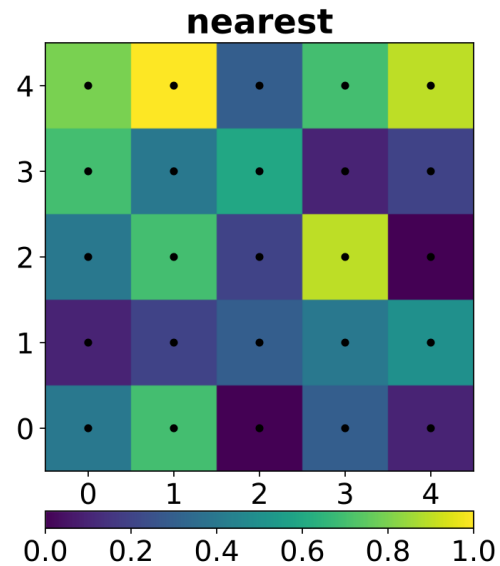
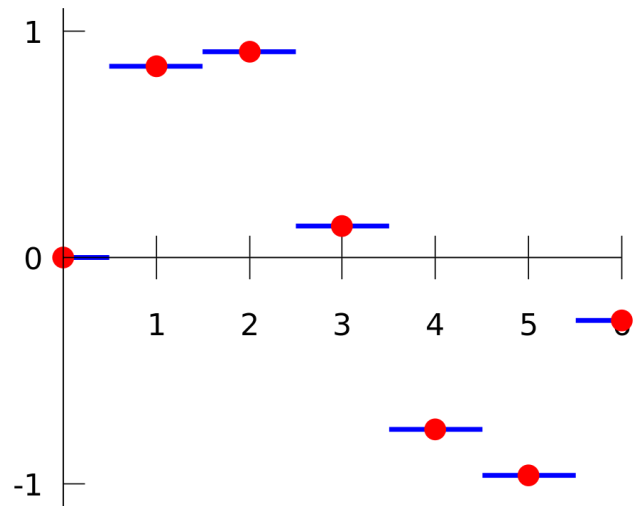
What should we do?

Intensity interpolation or **coordinate quantization**

- **Nearest neighbor Interpolation**
 - Choose the intensity of the closest pixel
- **Bilinear interpolation**
- **Bicubic interpolation**
- ...

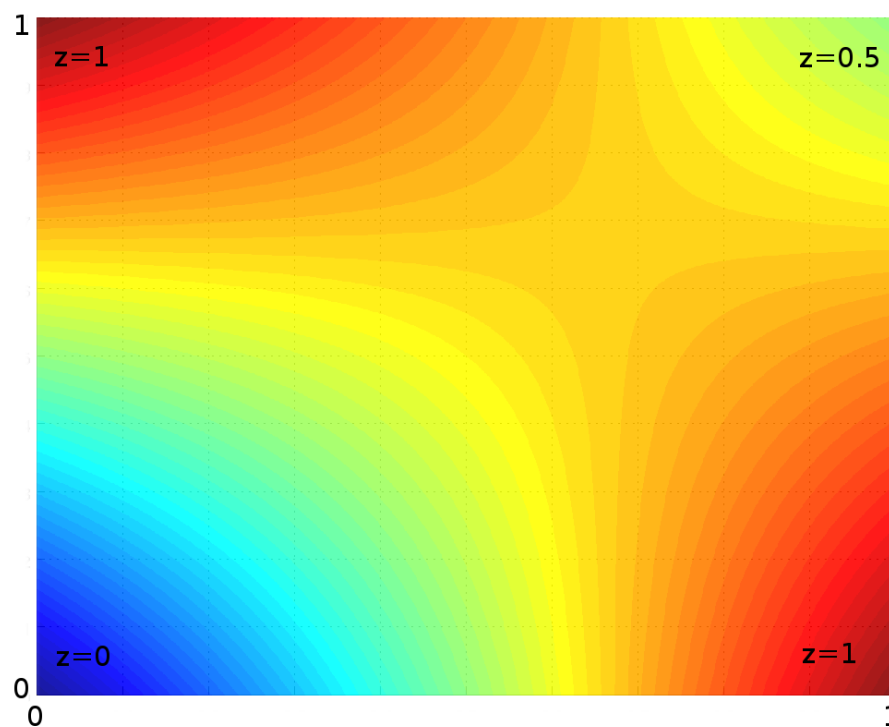
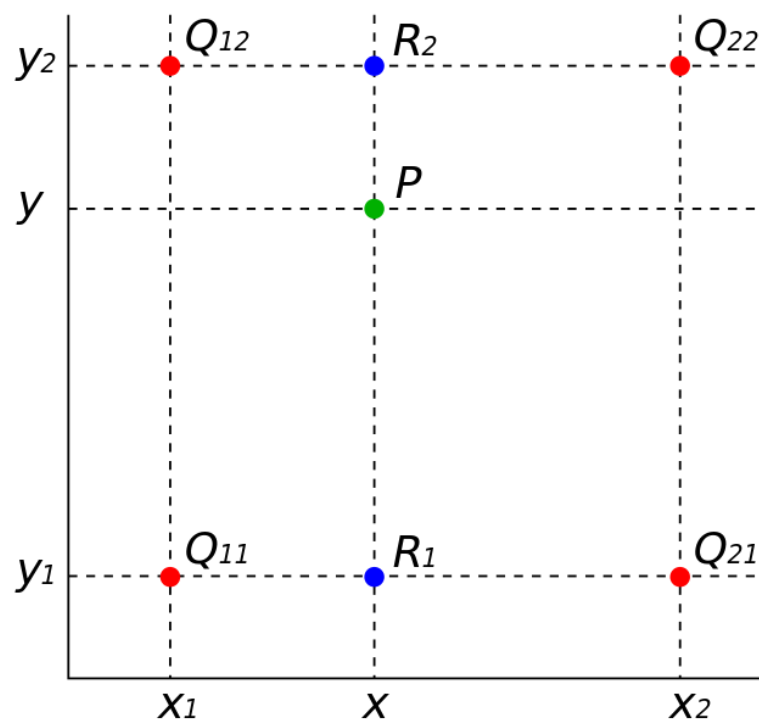
Interpolation

☐ Nearest Interpolation



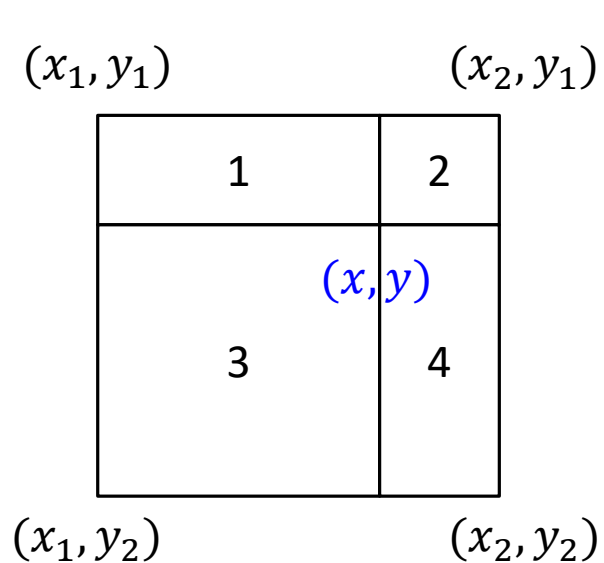


Bilinear Interpolation





Bilinear Interpolation



$$f(x, y) = \alpha f(x_1, y_1) + \beta f(x_2, y_1) + \gamma f(x_1, y_2) + \omega f(x_2, y_2)$$

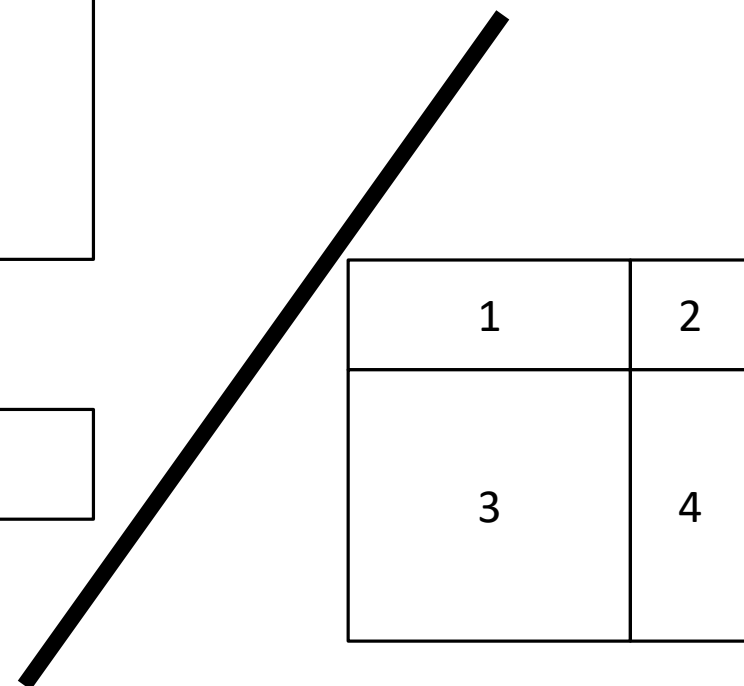
$$\alpha = \boxed{4}$$

$$\beta = \boxed{3}$$

$$\gamma = \boxed{2}$$

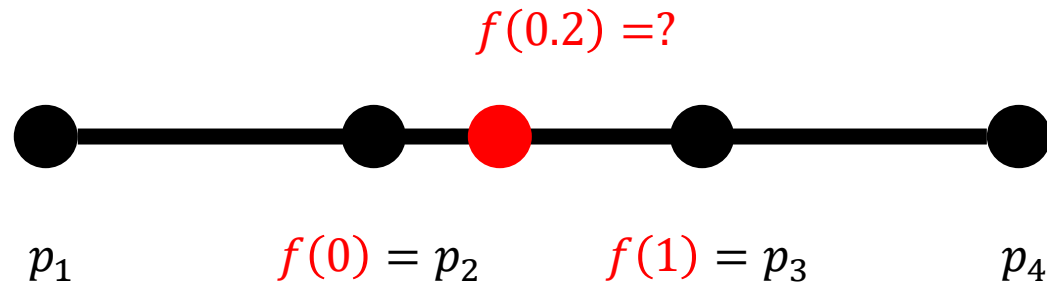
$$\omega = \boxed{1}$$

$$\alpha + \beta + \gamma + \omega = 1$$



Cubic Interpolation

- If we would like to interpolate a function $f(x)$ on the interval $[0,1]$ with its derivatives known at $x=0$ and 1 using a third degree polynomial, called cubic interpolation.



$$f(x) = ax^3 + bx^2 + cx + d \quad \longrightarrow \quad \begin{aligned} &\bullet f(0) = d, \\ &\bullet f(1) = a + b + c + d \\ &\bullet f'(0) = c \\ &\bullet f'(1) = 3a + 2b + c \end{aligned} \quad \longrightarrow \quad \begin{aligned} &\bullet a = 2f(0) - 2f(1) + f'(0) + f'(1) \\ &\bullet b = -3f(0) + 3f(1) - 2f'(0) - f'(1) \\ &\bullet c = f'(0) \\ &\bullet d = f(0) \end{aligned}$$

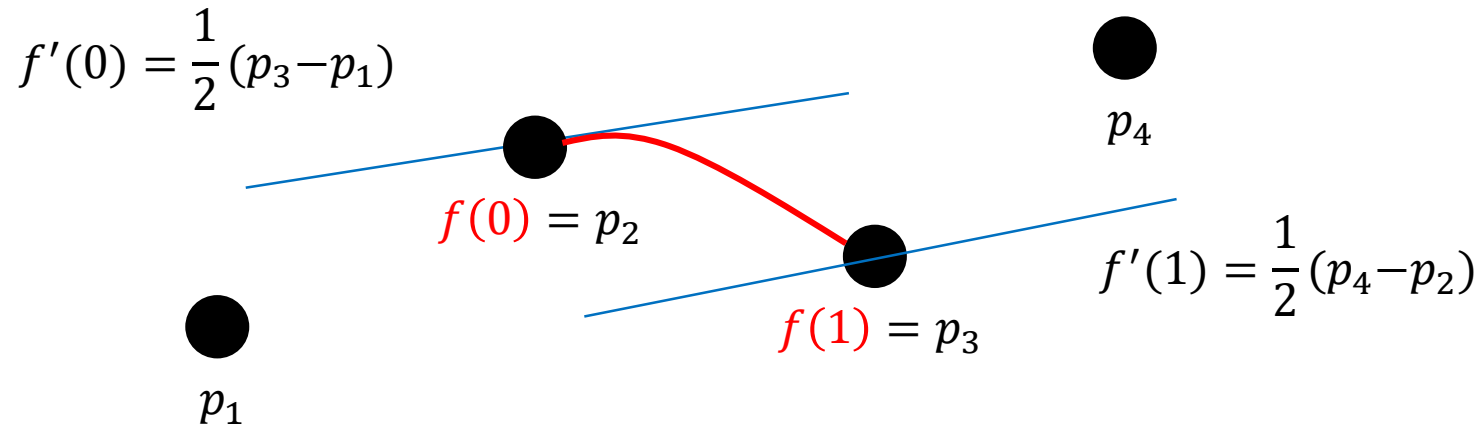
Cubic Interpolation

$$f(x) = ax^3 + bx^2 + cx + d \quad \longrightarrow$$

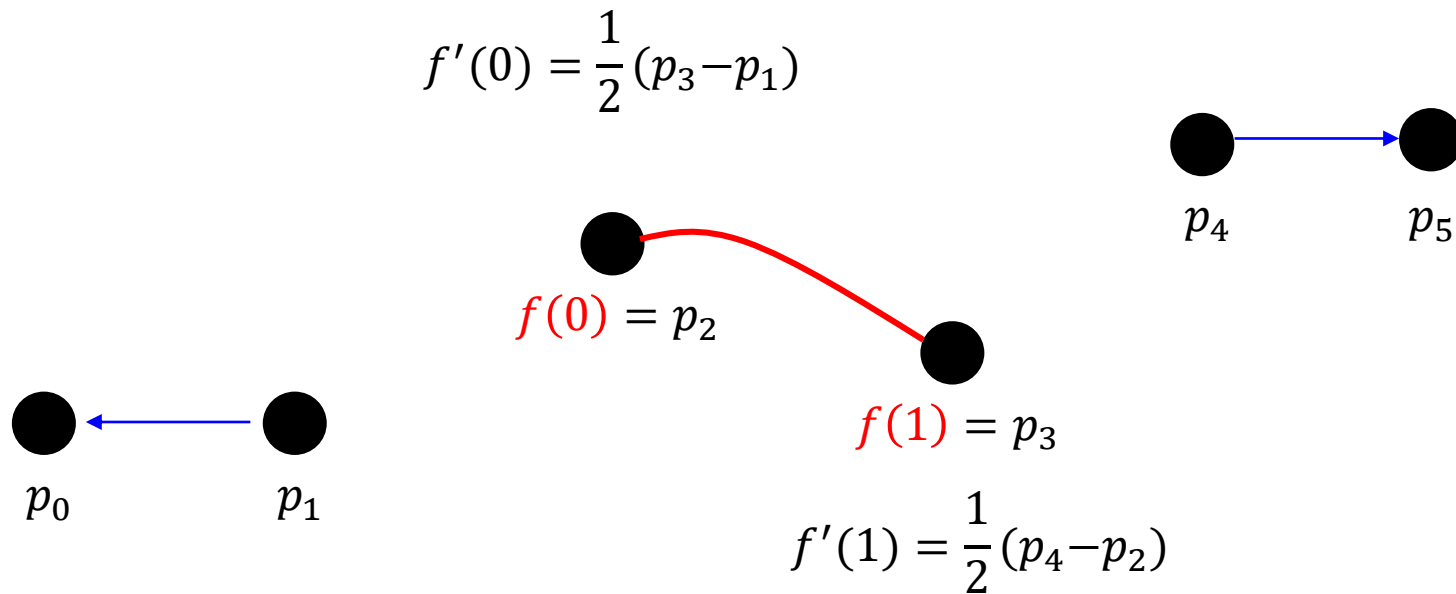
- $f(0) = d,$
- $f(1) = a + b + c + d$
- $f'(0) = c$
- $f'(1) = 3a + 2b + c$



- $a = 2f(0) - 2f(1) + f'(0) + f'(1)$
- $b = -3f(0) + 3f(1) - 2f'(0) - f'(1)$
- $c = f'(0)$
- $d = f(0)$

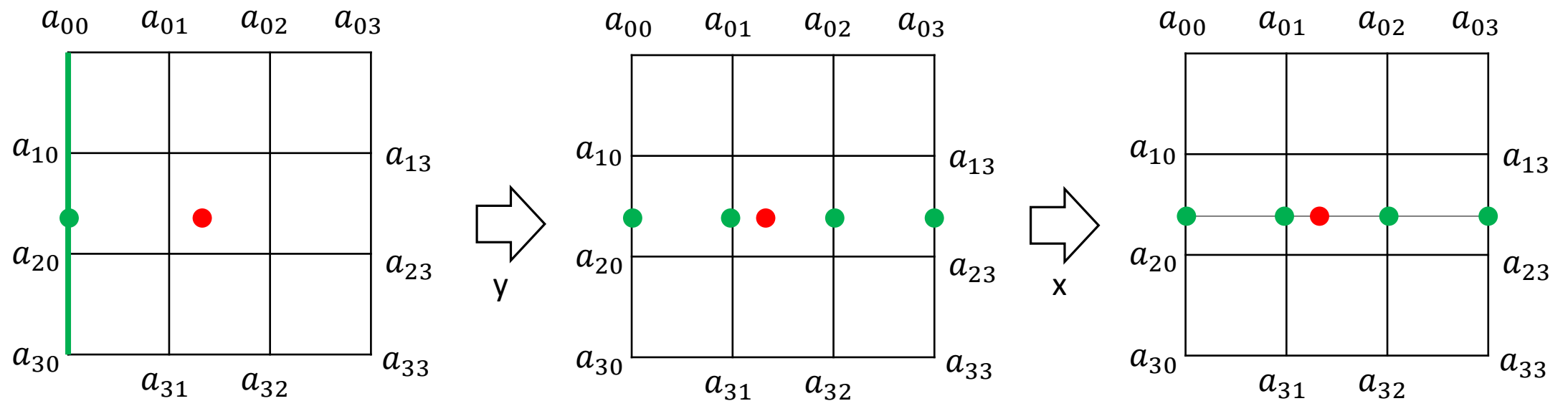


Cubic Interpolation



Bicubic Interpolation

- ❑ Do cubic interpolation twice
 - ❑ First in y-direction, and then in x-direction



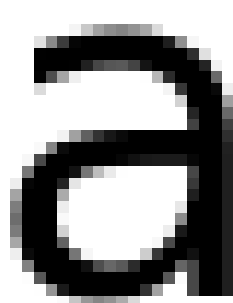
OpenCV Functions

❑ Resize

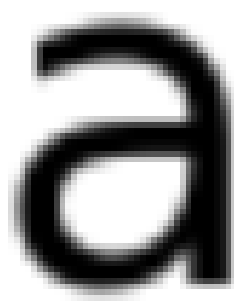
- ❑ `cv2.resize(src, dsize, interpolation mode)`
- ❑ `img = cv2.resize(img, (width, height), interpolation = cv2.INTER_CUBIC)`
- ❑ Interpolation mode: `INTER_NEAREST`, `INTER_LINEAR`, `INTER_CUBIC`

a

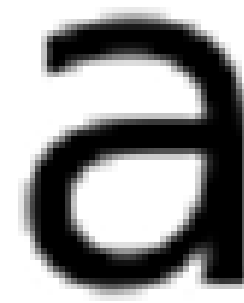
Original Image



INTER_NEAREST

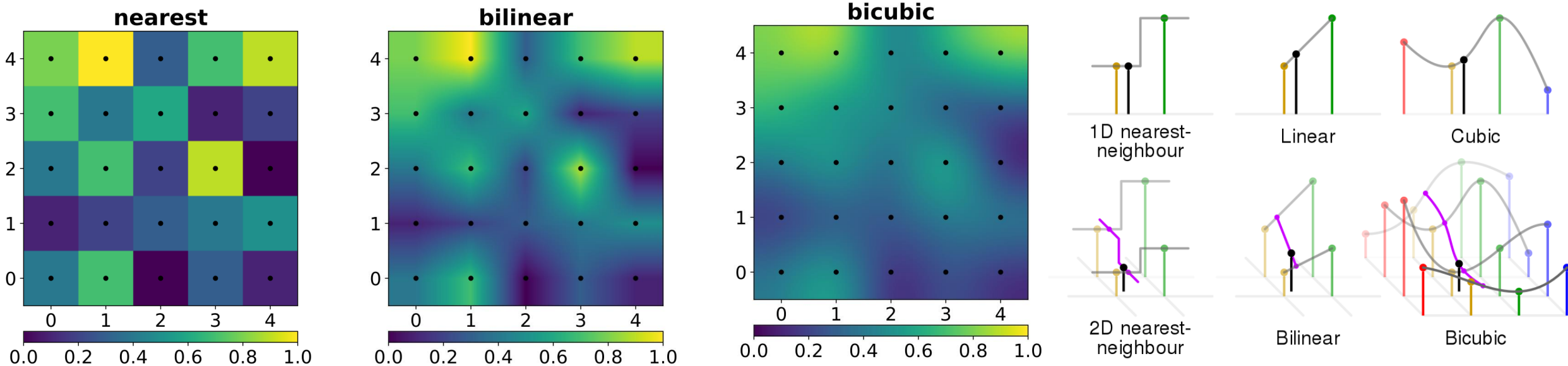


INTER_LINEAR



INTER_CUBIC

Comparison among Interpolations



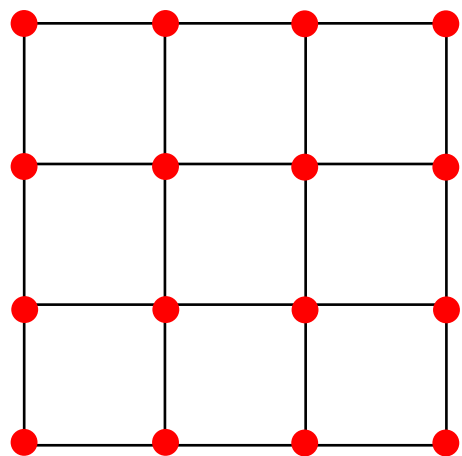
Forward Warping

Forward warping 的缺點：
可能產生「空洞 (holes)」：有些目標圖像的像素可能沒有被任何來源點對應到

需要插值處理 (resampling/interpolation)：把不規則分佈的黃點資訊「補回」到藍色網格中

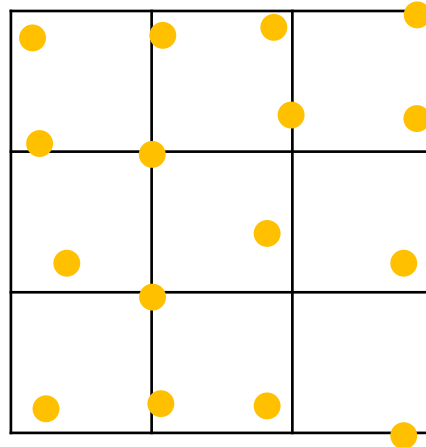
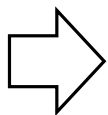
- ❑ To map every pixel in the source image to the target image

Reconstruct grid pixels from the irregular grid pixels



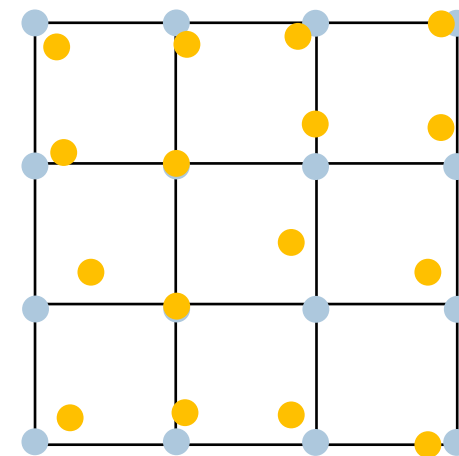
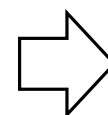
Source image

Original pixel



Target image

transformed pixel

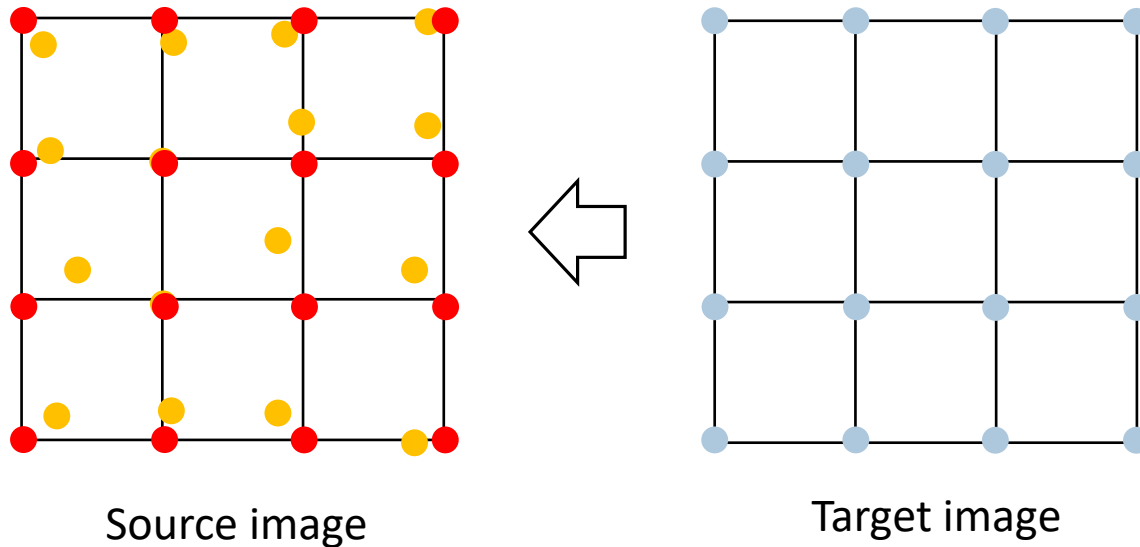


Actual wrapped pixel

每個目標圖像素都一定會被處理一次，不會漏掉任何像素，不會產生空洞

Backward(Inverse) Warping

- To map every pixel from the target image back to the source image

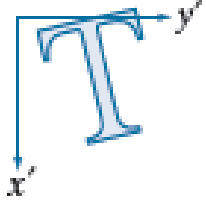
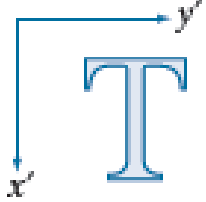
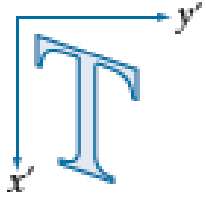
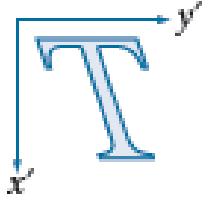


Suggestion: Use inverse warping

Forward warping could lead to missing pixels
(all the transformed pixels are too far away)

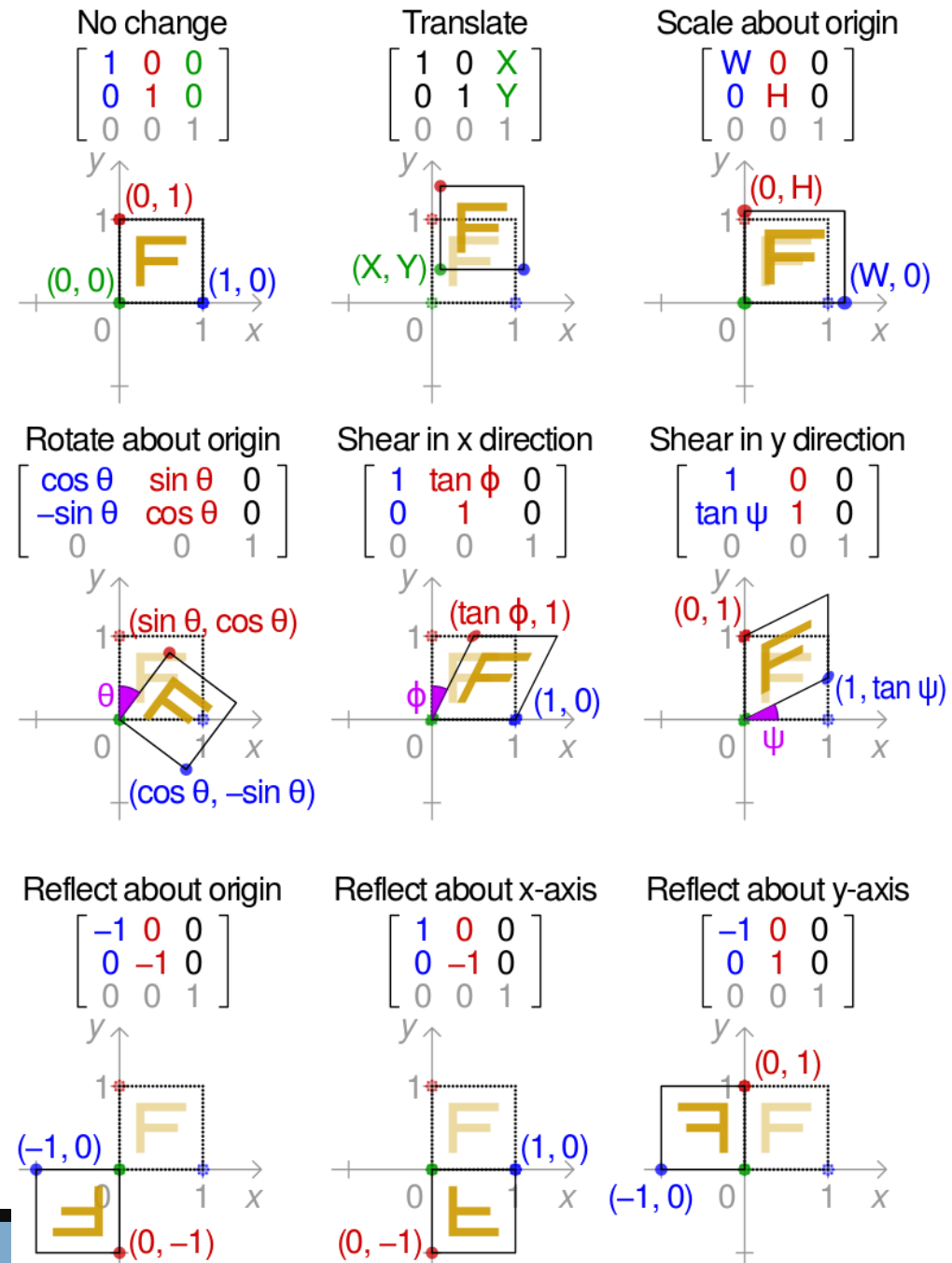
Reconstruct irregular grid pixels from the regular pixels

Geometric Transformations on Images

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Rotation (about the origin)	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x \cos \theta - y \sin \theta \\ y' &= x \sin \theta + y \cos \theta \end{aligned}$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x + t_x \\ y' &= y + t_y \end{aligned}$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x + s_v y \\ y' &= y \end{aligned}$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x \\ y' &= s_h x + y \end{aligned}$	

2D affine transformation matrices on a unit square

Affine transformation is defined as a linear mapping for spatial coordinate transformation, which preserves shapes, i.e., points, straight lines, and planes.



a	b
c	d

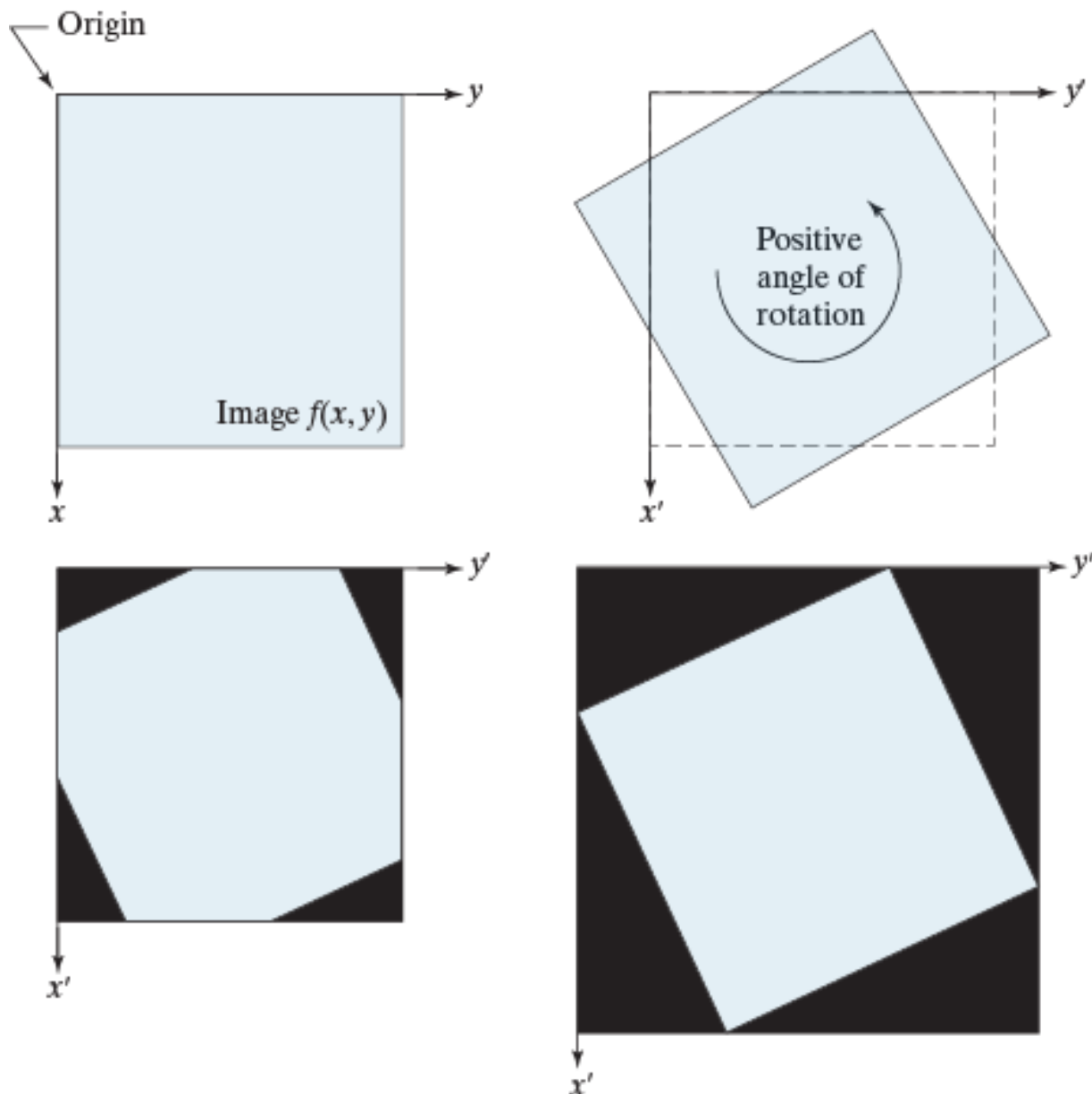
FIGURE 2.41

(a) A digital image.

(b) Rotated image (note the counterclockwise direction for a positive angle of rotation).

(c) Rotated image cropped to fit the same area as the original image.

(d) Image enlarged to accommodate the entire rotated image.



a	b
c	d

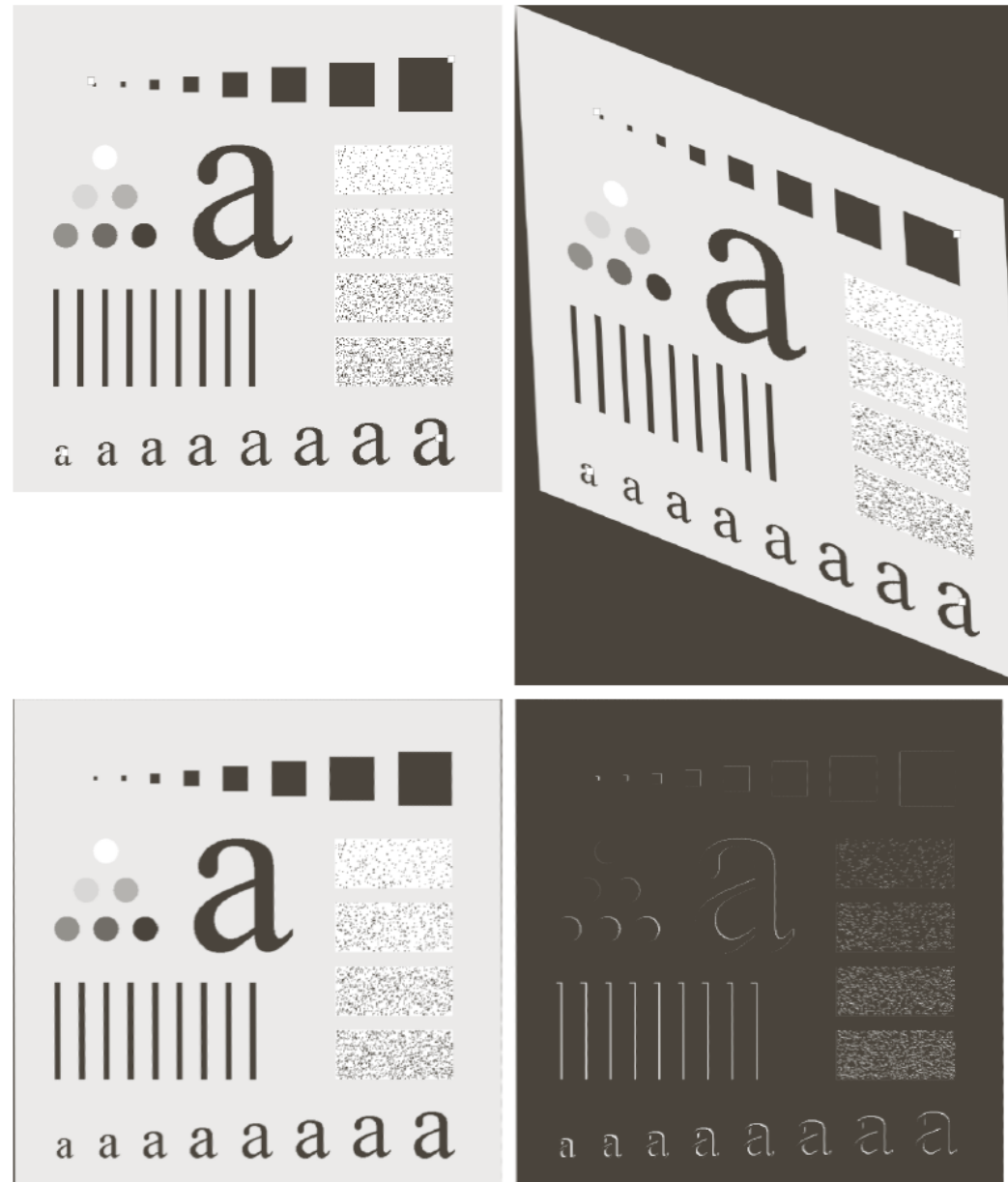
FIGURE 2.42

Image registration.

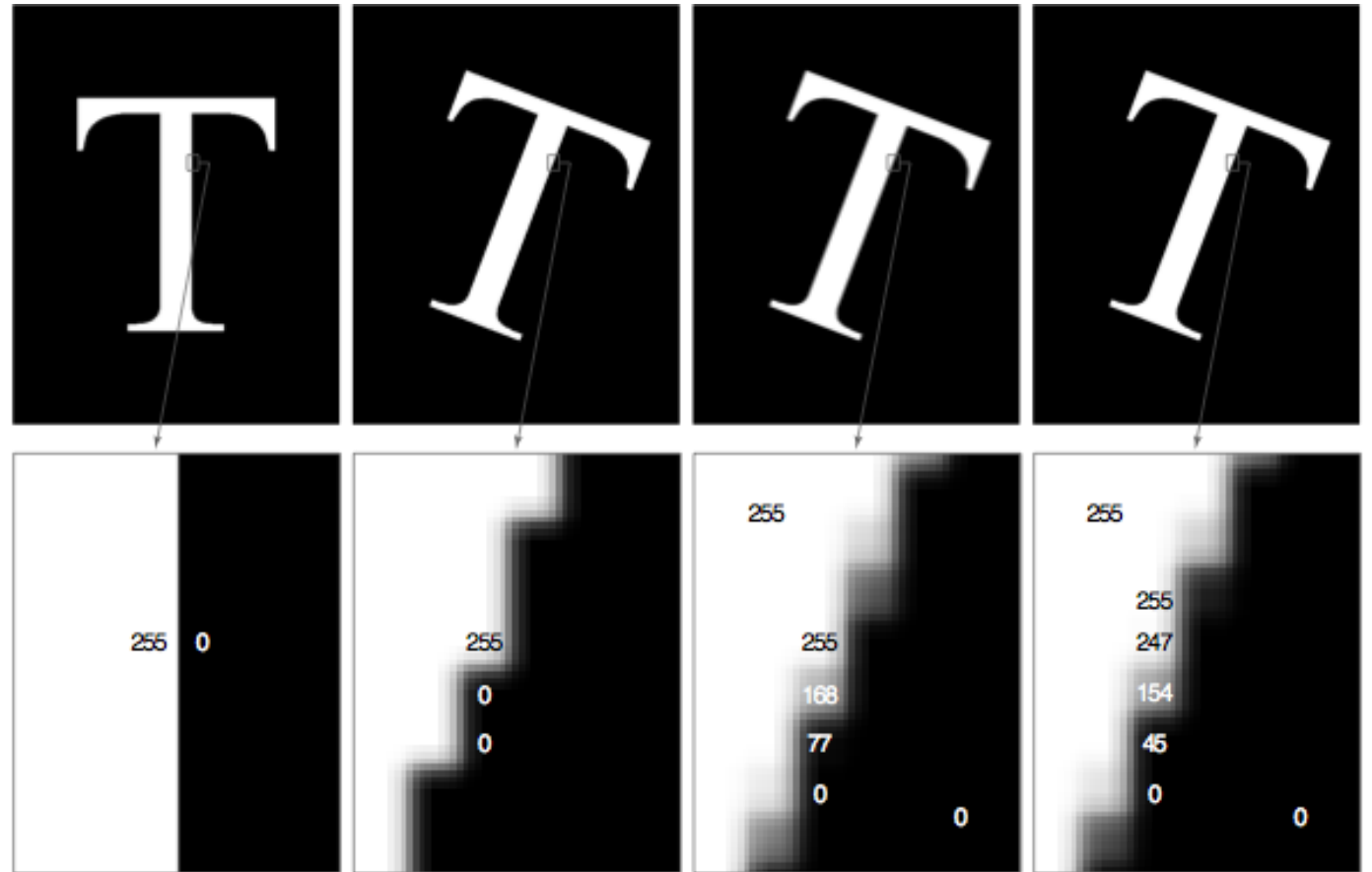
(a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners.

(c) Registered (output) image (note the errors in the border).

(d) Difference between (a) and (c), showing more registration errors.



- Interpolation occurs when you rotate or distort an image
- Rotation may reposition pixels onto the border



a b c d
e f g h

FIGURE 2.40 (a) A 541×421 image of the letter T. (b) Image rotated -21° using nearest-neighbor interpolation for intensity assignments. (c) Image rotated -21° using bilinear interpolation. (d) Image rotated -21° using bicubic interpolation. (e)-(h) Zoomed sections (each square is one pixel, and the numbers shown are intensity values).

OpenCV Functions – Affine Transformation

- ❑ Geometric Image Transformations
- ❑ `Img = cv2.warpAffine(src, M, dsize, interpolation mode, ..., borderValue=(r,g,b))`
 - ❑ `M` : transformation matrix
 - ❑ `dsize`: output image size
 - ❑ `borderValue`: filled background color
 - ❑ Rotation
 - ❑ Use `cv2.getRotationMatrix2D(center, angle, scale)` to get the rotation matrix
 - ❑ `center`: rotation center
 - ❑ `angle`: rotation angle
 - ❑ `scale`: size after rotation

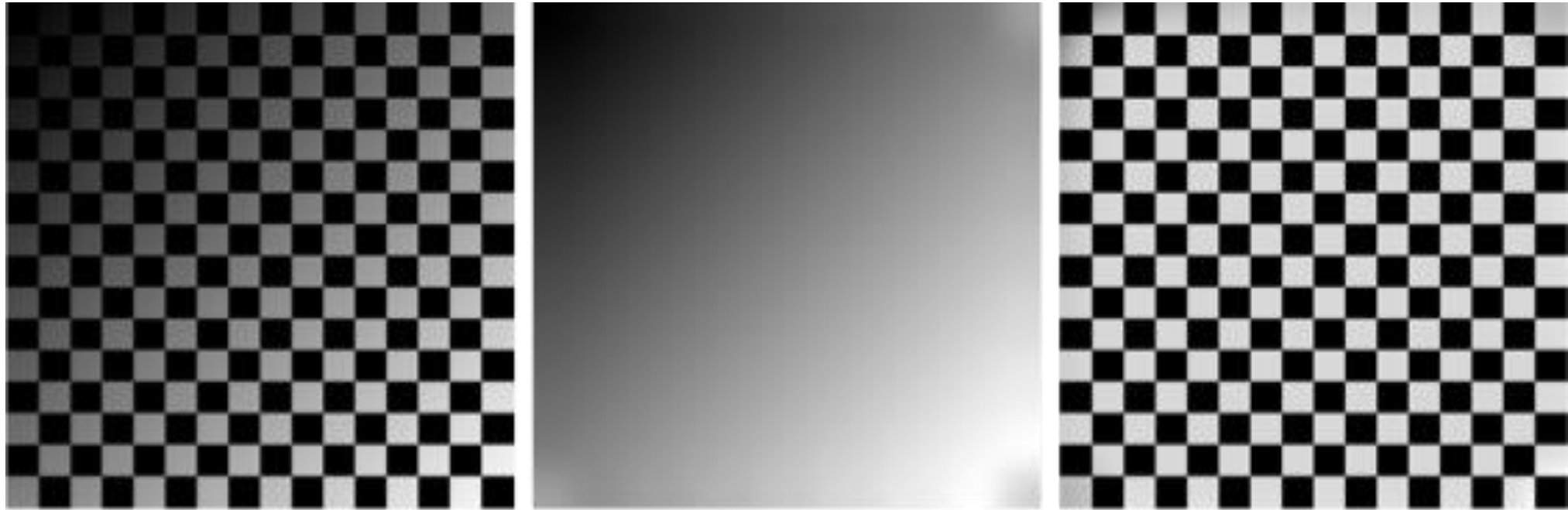
Lens Compensation

Geometric Transformation - Shear



Lens Shading

Array Product

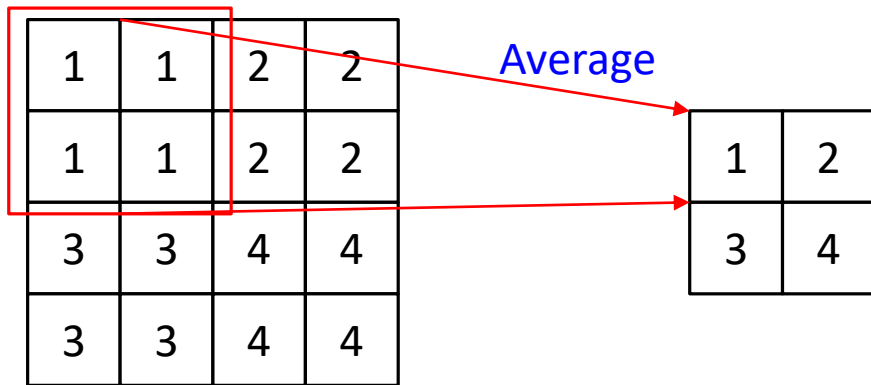


a b c

FIGURE 2.33 Shading correction. (a) Shaded test pattern. (b) Estimated shading pattern. (c) Product of (a) by the reciprocal of (b). (See Section 3.5 for a discussion of how (b) was estimated.)

Image Zoom-in and Zoom-out

Zoom-out



Zoom-in

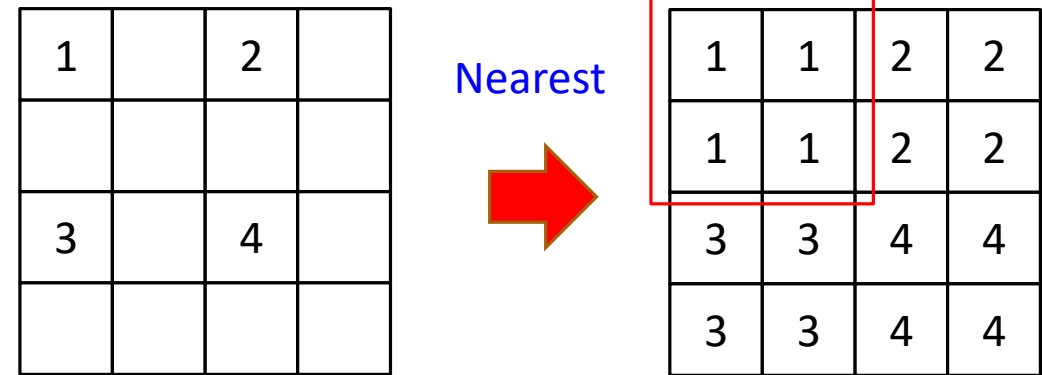


Image Transform

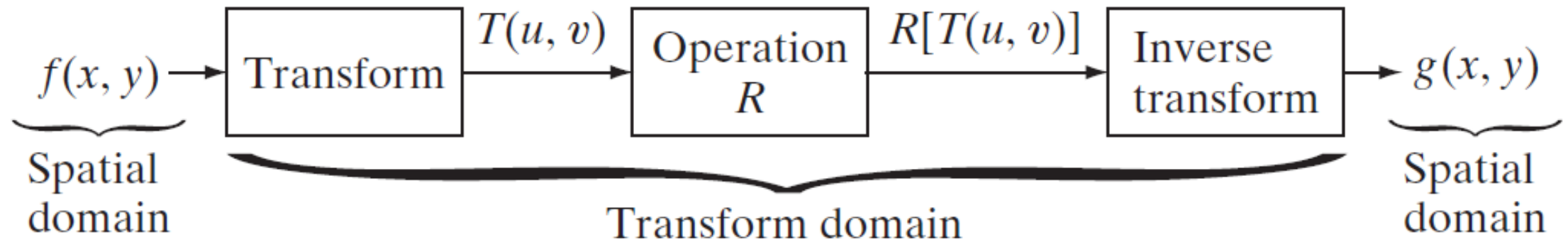
將空間影像轉成另一個表示方式 => 轉換域 (Transform Domain)

最常見的轉換：

Fourier Transform (頻率分析)

Discrete Cosine Transform (DCT) (影像壓縮)

Wavelet Transform (多解析度分析)



比如：

去除高頻 => 模糊

去除低頻 => 強化邊緣

壓縮小係數 => 壓縮影像

受到正弦波干擾，導致斜向條紋 (Moire pattern)

a b
c d

FIGURE 2.45

(a) Image corrupted by sinusoidal interference.
(b) Magnitude of the Fourier transform showing the bursts of energy caused by the interference (the bursts were enlarged for display purposes).
(c) Mask used to eliminate the energy bursts.
(d) Result of computing the inverse of the modified Fourier transform.
(Original image courtesy of NASA.)



頻率域

Fourier transform

Spatial Domain -> Frequency Domain

